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Survey on AI-Enabled Computer Vision Technologies and Applications for Space Robotic Missions

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ABSTRACT

This survey provides a comprehensive overview of recent advancements and challenges in Artificial Intelligence (AI)-enabled computer vision (CV) techniques for space robotic missions, spanning critical phases such as Entry, Descent, and Landing (EDL), orbital operations, and planetary surface exploration. Emphasis is placed on deep-learning-based approaches for image classification, object detection, semantic segmentation, relative pose estimation, and feature matching. State-of-the-art methods in terrain-relative navigation, crater-based or rock-feature matching, and pose estimation for uncooperative targets are highlighted, illustrating the progress achieved through hybrid pipelines combining deep neural networks with classical geometry. The paper also critically evaluates publicly available orbital and planetary data sets—along with the increasing role of synthetic data—for developing and benchmarking CV algorithms under strict resource limitations and harsh environmental conditions. Despite demonstrated success in tasks like autonomous landing, debris removal, and rover navigation, current solutions face significant hurdles. These include computational constraints on onboard hardware, insufficient coverage of planetary conditions in existing data sets, and limited adaptability to dynamically changing environments. To address these shortcomings, research must prioritize lightweight neural architectures, advanced synthetic data generation, adaptive or incremental learning, and robust multisensor fusion. By integrating these strategies, AI-based CV systems can advance autonomy, precision, and resilience in future space missions.

1 | Introduction

Computer vision (CV) has become a critical enabler of autonomous space robotic missions, providing unique advantages over traditional active sensors, such as radio detection and ranging (radar), Light Detection and Ranging (LiDAR), and laser rangefinders. Unlike these active sensors—which emit signals (e.g., electromagnetic waves or laser pulses)—CV systems operate passively using ambient light, thereby lowering power consumption, reducing hardware complexity, and minimizing

risks associated with signal interference. This benefit is especially important in resource-constrained environments. For instance, Mars rovers like Perseverance primarily rely on CV for terrain navigation while reserving radar for specialized tasks, such as subsurface ice detection. Although radar can provide robust long-range sensing and penetration of subsurface materials (e.g., detecting lunar ice), it typically requires larger antennas and higher power transmitters. LiDAR and laser rangefinders offer highly accurate depth information but depend on

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mechanically scanned laser emitters, which can become prone to mechanical failures or dust interference in planetary environments. Consequently, orbital servicers often prefer CV for proximity operations (e.g., debris removal), where LiDAR emissions could interfere with sensitive instruments.

In the last decade, Artificial Intelligence (AI) has played a pivotal role in advancing CV for intelligent systems, as evidenced by its adoption in commercial drones and electric vehicles. These successes have demonstrated the potential of AI-enabled CV to push the boundaries of fully autonomous systems. Correspondingly, space missions increasingly demand higher levels of autonomy, driving the need for CV pipelines capable of converting raw imagery into actionable insights—both for onboard decision-making and for ground-based operations. However, robust performance in AI-driven CV often hinges on large, high-quality training data sets. In space, such data are notoriously scarce, difficult to annotate, and narrowly focused on specific mission contexts. Consequently, it remains an ongoing challenge to collect sufficient labeled imagery to train convolutional neural networks and other deep-learning models for reliably handling extreme environmental conditions and specialized operational scenarios.

CV's passive operation, compact hardware, and flexibility make it particularly suitable for spaceflight applications. Radar systems often require large antennas and oscillators, and LiDAR depends on delicate mechanical components, whereas space-qualified cameras (e.g., those developed for European Space Agency's [ESA's] PILOT lunar lander) are relatively lightweight and radiation-hardened. This allows CV technologies to be readily incorporated into smaller satellites and rovers. Moreover, CV can thrive in conditions that challenge active sensors. For example, LiDAR struggles in dusty Martian environments, while the Rosalind Franklin rover's VisLoc system is designed to handle partial lens occlusions (Winter 2015). Conversely, radar remains indispensable for certain tasks—like detecting lunar ice beneath the surface—where visual data alone fall short.

Despite these advantages, deploying CV in space introduces formidable hurdles. First, data scarcity is universal: orbital and planetary imagery (e.g., for asteroid navigation in missions like Hayabusa2) is limited and often poorly labeled (Gaskell 2006). To mitigate this, some missions rely on synthetic imagery—such as NASA's Frontier Development Lab (FDL) Lunar Digital Elevation Models (DEMs)—and transfer learning. Second, extreme lighting variations, radiation, and dust can degrade sensor performance, prompting the use of multispectral or specialized imaging techniques. Third, limited onboard computing resources restrict the complexity of deep neural networks (DNNs), necessitating lightweight models (e.g., SqueezeNet) and hardware-accelerated algorithms. Real-time processing further intensifies these constraints, especially for mission-critical stages, like, Entry, Descent, and Landing (EDL). NASA's Terrain-Relative Navigation (TRN), for example, achieves real-time feature matching by downsampling orbital maps (Terrain Relative Navigation).

Space CV also differs substantially from terrestrial applications. In unstructured, low-feature environments—such as asteroid surfaces or the lunar regolith—techniques based on crater or landmark matching (Winter 2015) replace the road markings or other clear structures found on Earth. Communication delays

and the impossibility of human oversight in most space scenarios often demand fully autonomous operations. Sensor configurations also diverge: passive cameras predominate in space due to power constraints, whereas terrestrial robots often combine multiple active sensors (e.g., LiDAR and RGB-D) with extensive labeled data sets (e.g., Cityscapes and ImageNet). The cost and logistical difficulties of collecting comprehensive real-world imagery in space underscore the continued necessity of synthetic data and advanced AI techniques capable of addressing novel, data-limited conditions.

This survey consolidates current advances in AI-driven CV techniques for space robotic missions by dividing the discussion into six main sections. In Section 2, we outline the major challenges and state-of-the-art breakthroughs for three core mission phases—EDL, orbital operations, and planetary surface exploration. Section 3 provides a deeper look at key AI-based CV methodologies, such as image classification, object detection, semantic segmentation, relative pose estimation, and feature matching (Figure 1).

Building on these algorithmic foundations, Section 4 critically evaluates available orbital and planetary surface data sets, highlighting persistent gaps in labeled data and realism. Section 5 synthesizes the shortcomings that remain—especially computational constraints, domain adaptation, and limited onboard adaptability—and proposes concrete research directions for more robust and versatile CV systems. The paper is concluded in Section 6.

2 | Current Challenges and Advances in Space CV Technologies

2.1 | Entry, Descent, and Landing

EDL remains one of the most critical and challenging phases in space robotic missions, demanding high-accuracy navigation and robust autonomy to safely land on planetary bodies. EDL missions commonly rely on constructing or refining a landmark database throughout the mission, which contains georeferenced coordinates of morphological features (e.g., craters and ridges), along with their sizes. As new observations are collected, the database is updated to increase resolution and accuracy. Depending on downlink latency, onboard memory constraints, and available processing power, this database can be managed either on-ground or fully onboard.

2.1.1 | Key Challenges

- *Precise navigation and orientation:* Missions require absolute navigation to correctly align with the planetary reference frame. Landmark-based methods often hinge on an initial guess of the spacecraft's position and orientation, which is then refined by comparing real-time camera views with predicted landmark projections (Terrain Relative Navigation).
- *Environmental complexities:* Harsh conditions—such as extreme lighting, dust, or partial sensor occlusions—degrade visual performance and demand robust image processing (Mammarella et al. 2011).
- *Computational constraints:* Deep-learning or feature-based algorithms must be optimized to run in real time on limited



FIGURE 1 | Future AI and robotics hub for space (FAIR-SPACE).

onboard hardware, particularly during critical parachute descent phases.

2.1.2 | Recent Advancements

- **TRN:** NASA's Mars2020 Perseverance rover employs TRN by loading an orbital-derived map onboard, then capturing images during its descent phase and matching them to the stored map to pinpoint its location accurately ([Terrain Relative Navigation](#)). This real-time landmark comparison has substantially improved landing precision over previous Mars landers.
- **Crater detection and landmark matching:** ESA's PILOT system (slated for use on Luna-27) integrates crater detection and matching against a prestored database for precise absolute navigation relative to the lunar surface (Mammarella et al. [2011](#)).
- **Landing Map (L-map) approach for smaller bodies:** Missions like Hayabusa, Dawn, and Rosetta rely on three-dimensional (3D) "L-map" models of terrain patches, which are rendered to align with real images via intensity correlation. This approach does not depend on obvious morphological features like craters, making it especially effective for asteroid or comet landings (Gaskell [2006](#); Mastrodemos et al. [2011](#)). It is also part of the target's characterization and mapping pipeline, using stereo-photoclinometry to generate a global shape model ([Stereo-photoclinometry](#); Gaskell [2008](#)).

2.1.3 | Limitations and Open Challenges

- **Reliance on prior data:** TRN, PILOT, and L-map techniques all require high-quality, preexisting topographic data or shape models, reducing effectiveness if the target body is not well-studied or experiences dynamic changes (Gaskell [2006](#); Mammarella et al. [2011](#)).

- **Limited adaptability:** Even with robust models, unanticipated changes to terrain morphology, extreme lighting, or dust storms can threaten the reliability of EDL systems (Gaskell [2006](#); Mammarella et al. [2011](#); Mastrodemos et al. [2011](#)).

- **Onboard-resource trade-offs:** Generating and matching terrain models in real time is computationally expensive, forcing system designers to downsample maps or limit update frequency, potentially impacting precision (Winter [2015](#)).

2.1.4 | Future Research Directions

- **Adaptive, real-time CV:** Development of more flexible algorithms capable of reacting to unforeseen changes in the landing environment without extensive reliance on ground updates.
- **Lightweight deep-learning architectures:** Improving the performance-per-watt efficiency of neural networks, thus balancing real-time demands with limited onboard computation.
- **Robust multisensor fusion:** Enhanced integration of vision data with other sensors (radar, LiDAR, and inertial measurement units [IMUs]) to mitigate single-sensor vulnerabilities and strengthen overall robustness (Winter [2015](#)).

Incorporating these improvements should ensure safer and more adaptive EDL operations, ultimately enabling more ambitious deep-space exploration and robotic science missions.

2.2 | Orbital Operations

Orbital operations demand precise vision-based Guidance, Navigation, and Control (GNC) for tasks, such as rendezvous, docking, active debris removal, satellite assembly, refueling, maintenance, and manufacturing. Achieving long-term

sustainability and exploitation of space hinges on robust AI-enabled perception systems capable of handling minimal human intervention and unpredictable orbital conditions.

These mission objectives impose challenging requirements on vision-based GNC, where the relative position and attitude of one spacecraft with respect to another must be accurately determined for tasks like rendezvous or docking, or where the vehicle's position must be tracked against extraterrestrial bodies during exploration. Such requirements underscore the need for advanced, real-time perception algorithms that can adapt to the dynamic nature of orbital environments.

2.2.1 | In-Orbit Servicing and Manufacturing

In-orbit asset assembly, refueling, maintenance, and repair will be key for sustainable space exploration and commercialization. Unlike prescribed missions, many GNC tasks in these scenarios must be decided on the fly, relying on minimal ground support. Advanced vision-based perception and AI methods, including robust object detection and pose estimation, are crucial:

- *Autonomous robotic payloads:* Onboard manipulators or grippers depend on precise CV for situational awareness, object tracking, and dexterous interaction with target spacecraft components (Gao 2021; Mavrakis and Gao 2020; Lee et al. 2018).
- *Modeling and simulation:* AI-based systems can integrate with multiphysics and multiscale simulations, enabling validation and optimization of in-orbit servicing algorithms before deployment. This synergy further reduces mission risk and cost.
- *Cost-effectiveness:* Inexpensive cameras combined with refined CV pipelines can offer a potent, low-power approach to gathering relative pose information, reducing reliance on more expensive sensor suites (Mavrakis and Gao 2020).

2.2.1.1 | State-of-the-Art Advancements

- *Vision-based relative navigation:* Missions leveraging in-orbit servicing concepts often use camera-based navigation solutions that integrate AI-driven algorithms for robust relative pose determination (Gao 2021; Mavrakis and Gao 2020; Lee et al. 2018).
- *Multisensor data fusion:* Data from cameras, LiDAR, and other sensors (e.g., thermal imagers) are fused to improve the reliability of object identification and robotic manipulations (Mavrakis and Gao 2020; Lee et al. 2018).
- *AI-based grasp synthesis:* Ongoing research explores using Convolutional Neural Network (CNN)-based methods to determine stable grasps of noncooperative targets, such as rocket stages or defunct satellites (Gao 2021; Mavrakis and Gao 2020; Lee et al. 2018).

2.2.1.2 | Persistent Limitations

- *Adaptive real-time control:* Onboard GNC must respond quickly to dynamic interactions between robotic arms and target structures, requiring efficient AI.

- *Algorithm robustness:* Reflective spacecraft surfaces and partial occlusions can degrade CV performance.
- *Computational constraints:* Resource-heavy algorithms require significant optimization for real-time operation on flight-grade processors.

In Table 1, heavy models (e.g., DeepLabv3+) exceed spacecraft compute budgets, confirming that lightweight architectures like YOLOv3-tiny and MobileNet are necessary for onboard deployment.

2.2.1.3 | Future Research Directions

- *Optimized AI architectures:* Development of lightweight, robust deep-learning models capable of running on limited onboard hardware.
- *Virtual testing environments:* Comprehensive multiphysics simulations help refine servicing approaches before real deployments.
- *Enhanced autonomy:* Incorporating online adaptation strategies (e.g., reinforcement learning) for midmission scenario changes.

2.2.2 | Small Bodies Reconnaissance

Reconnaissance of small bodies (e.g., asteroids and comets) is essential for scientific discovery and resource prospecting. High-quality vision-based navigation and robust autonomy are pivotal for proximity operations:

- *Close-proximity maneuvers:* Missions such as ESA's AIM and HERA rely on vision-based GNC in concert with additional sensors (e.g., LiDAR) to accurately estimate spacecraft states at close ranges, facilitating safe, precise orbits, or fly-bys (Gaskell 2006; Mastrodemos et al. 2011). A precise and robust on-board navigation estimation of the spacecraft state is key to progressively reducing the pericenter of these maneuvers—sometimes to just a few hundred meters—while mitigating collision risks.
- *Contingency handling:* An autonomous GNC system constantly monitors for unexpected maneuver-execution errors or trajectory deviations. In case of failures (e.g., unpredicted maneuvers leading the spacecraft into a collision course), the system must trigger Collision Avoidance Maneuvers to ensure safety.
- *Case study—Comet Interceptor:* A future ESA-JAXA mission aimed at intercepting a Long Period Comet, a Dynamically New Comet, or an interstellar body in a fly-by scenario. The Comet-I mission faces unique challenges due to limited a priori knowledge of the target's morphology and environment, necessitating image-processing pipelines that blend classical techniques (e.g., centroiding on brightest pixels) with covariance-based jet compensation. Common model-matching algorithms may be inapplicable if the comet's characteristics are too uncertain.

2.2.2.1 | State-of-the-Art Advancements

- *Vision-based relative navigation:* ESA's AIM mission demonstrated effective vision-based proximity maneuvers

TABLE 1 | Computational load versus spacecraft hardware constraints.

Model	Parameters (M)	FLOPs (G)	Inference latency (ms)	Typical platform	Onboard feasibility/constraint	Cited paper(s)
DeepLabv3+	44	568.0	~625	Qualcomm Adreno 660 GPU (Samsung Galaxy S21)	Exceeds real-time constraints (> 100 ms)	L.-C. Chen et al. (2018), Swan (2021), and Y. Li et al. (2023)
Lightweight CNN (e.g., MobileNet family)	~2.5–3 after pruning/optimization	~0.15–0.3	—	LEON4 CPU (≤ 10 W)	Feasible for onboard inference	Luo et al. (2020)
YOLOv3-tiny	~8.7–9.0	~2.8	~22	FPGA/embedded edge accelerator (~2–3 W)	Meets real-time constraints (< 20–30 ms)	An et al. (2025) and Y.-M. Zhang, Lee et al. (2021)

Abbreviations: CNN, Convolutional Neural Network; CPU, Central Processing Unit; FLOPs, floating point operations per second; FPGA, Field-Programmable Gate Array; GPU, Graphics Processing Unit; YOLO, You Only Look Once.

around near-Earth asteroids, complemented by sensor fusion (e.g., data fusion with a laser altimeter). The HERA mission extends these concepts for improved precision when flying close to small bodies (Gaskell 2006; Mastrodemos et al. 2011).

- *DNN-based pose estimation:* The ESA Spacecraft Pose Estimation Challenge advanced CNN-based frameworks (e.g., Spacecraft Pose Network [SPN] and keypoint regression) for cooperative and noncooperative targets (Sharma and D'Amico 2019; B. Chen et al. 2019; Castano et al. 2003; Gerard 2019; Proenca and Gao 2020; Dhamani et al. 2020; Harvard et al. 2020; Rathinam and Gao 2020). Although originally benchmarked on orbital scenarios, similar AI-driven approaches stand to benefit small-body reconnaissance, provided they address the irregularities of asteroid/comet surfaces and harsh environmental conditions.

2.2.2.2 | Persistent Limitations

- *Algorithmic fragility:* Unknown or highly irregular morphology, extreme lighting, and dust can confound even advanced CNNs, leading to inaccurate pose solutions in close-proximity scenarios.
- *Real-time performance:* Rendering high-fidelity environment models or running deep CNNs at flight rates is difficult on resource-limited spacecraft hardware.
- *Generalization barriers:* Many systems are tuned for a specific mission (e.g., AIM/HERA), restricting adaptability to comets or asteroids of significantly different composition or shape.

Table 2 quantitatively illustrates the main limitations of small-body reconnaissance systems. It shows how CNN-based pose estimators struggle with lighting and shape variability, fail to achieve real-time rates on flight hardware, and generalize poorly to new targets. The results highlight the need for lighter and more adaptable onboard models.

2.2.2.3 | Future Research Directions

- *Domain adaptation:* Generating realistic synthetic data, combined with advanced transfer learning, can improve algorithm resilience for unfamiliar surfaces and dynamic lighting conditions.
- *Lightweight onboard inference:* Specialized hardware accelerators or compressed neural networks are crucial to achieving near real-time performance under strict power and computational budgets.
- *Autonomous hazard management:* AI-driven hazard detection and collision avoidance techniques reduce reliance on ground intervention, vital for missions operating near poorly characterized or unpredictable small-body targets.

2.2.3 | Common Limitations and Prospects for All Orbital Operations

Though servicing/manufacturing missions and small bodies reconnaissance present some unique hurdles, they share several overarching challenges and potential growth areas:

TABLE 2 | Quantitative illustration of persistent limitations.

Limitation	Representative method/mission	Accuracy or pose error	Inference rate/latency	Platform/hardware	Observation/constraint	Cited paper(s)
Algorithmic fragility	CNN+ line-segment pose estimator	1.9 cm, 0.73° (best-case); > 4× error under lighting/occlusion	~95 ms/frame	Jetson TX2 (7.5 W)	Accuracy drops > 4× under poor lighting or shadows	Bechini et al. (2024)
Real-time performance	Edge-TPU pose CNN	cm-level/subdeg accuracy	130 ms/frame (7.7 FPS)	Coral Dev Board Mini (2.2 W)	Below 30 Hz requirement for real-time control	Lotti et al. (2022)
Generalization barriers	Domain-adapted CNN for SPEED v2	2.5 cm/1.1° (mean); baseline > 6 cm	≤ 40 ms/frame (25 FPS)	Jetson Nano (4 GB)	Accuracy on unseen targets and glare	T. H. Park and D'Amico (2025)

Abbreviations: CNN, Convolutional Neural Network; FPS, frames per second; SPEED, Spacecraft Pose Estimation Dataset; TPU, thermoplastic polyurethane.

- *Algorithmic robustness:* The CV pipelines must cope with changing lighting, reflectivity, occlusion, and potential sensor degradation over long missions.
- *Real-time, power-efficient processing:* Future hardware design and specialized AI accelerators can help.
- *Scalability and generalization:* More flexible, modular solutions that adapt across diverse orbital contexts will improve mission success rates.

2.3 | Planetary Surface Exploration

Planetary surface exploration necessitates advanced vision-based technologies to enable autonomous navigation, mapping, and situational awareness under harsh environmental conditions and communication delays with Earth. Rovers and landers must independently plan and execute traverses, handle uncertainties, and make timely decisions, all while operating on severely limited hardware resources.

To provide the situational awareness and context for ground-based planning as well as to support on-board autonomous navigation and decision-making, significant resources have been applied to the development and qualification of visual sensors for planetary exploration. Navigation based on vision has always been an attractive option from a system-level perspective for missions, owing to the passive mode of operation, simple hardware, small form factors, and low power consumption of camera sensors. However, the deployment of on-board autonomous navigation based on vision has historically been limited, as these methods are typically, highly, computationally demanding, a particularly challenging obstacle, given the constraints associated with available computing resources.

It is largely for this reason that Earth-based activity planning has remained the default Concept of Operations applied to all exploration missions to date. Future rovers, however, will have to overcome this limitation to achieve evermore ambitious mission requirements, such as traversing further in a shorter time and at faster speeds, as well as increasing the number and complexity of operations they can perform autonomously using onboard processing. Missions currently driving these requirements include Sample Fetch Rover (SFR), as part of Mars Sample Return (MSR), which seeks to traverse up to 2 km of map distance in a maximum duration of 92 sols, covering up to 300 m/sol (Delfa Victoria et al. 2022). Burrowing and subsurface sampling mechanisms remain key challenges in planetary surface missions; recent reviews, such as Wei et al. (2021) summarize the state of bioinspired regolith-burrowing robots and highlight design trade-offs under low gravity and granular media. Additionally, the main scenarios currently considered for lunar exploration, as part of the on-going European Moon Rover System Pre-Phase A study—the Polar Explorer (ESA Polar Explorer CDF Report, CDF-212(A)), In Situ Resource Utilization (ISRU) Plant (ISRU Pilot Plant CDF Report, CDF-182(A)), and Astrophysics Lunar Observatory (Astrophysics Lunar Observatory CDF Presentation)—introduce various additional technical challenges to be addressed through the use of onboard autonomy. This likely will see solutions founded on CV in terms of visual localization, mapping, and terrain classification.

2.3.1 | Key Challenges

- *Real-time autonomy requirements:* Rovers must navigate with minimal ground intervention due to communication latencies and limited bandwidth (Y. Cheng et al. 2005).
- *Complex and unstructured terrains:* Surface environments can feature highly varied geological formations, loose soil, and dust storms that complicate navigation (Rankin et al. 2021).
- *Computational limitations:* Space-qualified hardware often operates at lower speeds and memory capacities than terrestrial systems, demanding highly optimized CV algorithms (Winter 2015).
- *Localization drift and map uncertainty:* Long-duration missions require precise localization to mitigate cumulative drift errors and maintain navigation accuracy over extended distances (Weclewski et al. 2025).
- *Limited labeled data sets:* Training AI models for planetary exploration is hindered by the scarcity of annotated planetary image data (Swan 2021; Barrett et al. 2022; Ono 2016; Palafox et al. 2017; Rothrock et al. 2016; DeLatte et al. 2019; Tewari et al. 2022; Silburt et al. 2019; Pohlen et al. 2016; M. Yang et al. 2018; J. Long et al. 2015; Zhou et al. 2020; Goh et al. 2022; Song and Shan 2008; Ono et al. 2015; Furlán et al. 2019; Kuang et al. 2021).

2.3.2 | Localization or Navigation

Rover localization (or navigation in GNC terms) focuses on accurately maintaining an estimate of the rover's position and orientation relative to planetary surfaces. Traditional techniques often relied on periodic ground updates, but the drive for greater autonomy has elevated the importance of onboard visual navigation:

- *Stereo Visual Odometry (VO):* NASA's Mars Exploration Rovers (MERs) demonstrated the feasibility of VO on a 20-MHz Central Processing Unit (CPU) (Cheng et al. 2005). Later missions (e.g., Mars Science Laboratory and ExoMars) incorporated faster flight processors (e.g., RAD750 and LEON2) to achieve improved odometry fidelity (Winter 2015; Rankin et al. 2021; Bora et al. 2017). In Europe, ESA's Rosalind Franklin rover hosts stereo-based VO (VisLoc) on a dedicated 96-MHz LEON2 co-processor, combining wheel odometry and gyro data to enhance relative position estimation (Winter 2015).
- *Absolute positioning updates:* Precise visual localization is critical for targeted subsurface penetration, as in bioinspired burrowing systems discussed by Wei et al. (2021). Despite robust relative localization, rovers occasionally require global alignment. The ExoMars rover's Absolute Localization (AbsLoc) module compares onboard accelerometer data and mission-related knowledge (e.g., sun sensing or ground updates) to maintain an absolute attitude (Bora et al. 2017). The SFR extends these concepts using DEMs and orthoimages for Absolute Global Localization (AGL) to reduce accumulated drift (Weclewski et al. 2025).
- *Hybrid localization methods:* Combining Monte-Carlo Localization (MCL), deep learning, and VO can align

onboard imagery with orbital maps, achieving more accurate global and local state estimates (Van Pham et al. 2003; Geromichalos et al. 2020; Dinsdale et al. 2022; Sheshadri et al. 2012; Hwangbo et al. 2009; Boukas et al. 2018; C.-S. Chen et al. 1999; Naguib 2018; Chung et al. 2018a; Wu 2019; Franchi and Ntagiou 2021; Matthies et al. 2022).

- *Salient feature-based matching:* Advanced feature-matching techniques leveraging DNNs such as Siamese Neural Networks (SNNs) have been employed to match rover images with orbital orthoimages, improving global localization accuracy (Chung et al. 2018a; Wu 2019).

2.3.3 | Mapping

Accurate environmental mapping is critical for safe, efficient path planning, and situational awareness on planetary surfaces. Modern approaches exploit binocular stereo inputs, constructing and updating local and regional terrain models:

- *Multiresolution stereo mapping:* ESA's Rosalind Franklin rover merges multiple disparity maps gathered at successive stops to produce DEM products. A statistical model estimates the mean, minimum, and maximum elevation for each cell, crucial for hazard avoidance and traversability analysis (Winter 2015).
- *Real-time feasibility:* Due to hardware restrictions, rover stereo processing often employs a multiresolution approach, limiting resolution while preserving coverage. Processing time must be minimized (e.g., < 20 s per navigation stop) (Winter 2015).
- *SPARTAN and space-sweep algorithms:* Development activities, such as ESA's SPARTAN, investigate dense stereo reconstruction methods (e.g., space sweep) that directly compute depth without relying on conventional disparity maps. These techniques exploit parallelism and allow modulating computation for anytime performance SPARTAN (Avilés et al. 2018).
- *Feature-based topographic matching:* Deep-learning-driven feature matching between orbital and rover-acquired DEMs helps refine terrain maps and improve rover localization (Van Pham et al. 2003; Geromichalos et al. 2020; Dinsdale et al. 2022).

2.3.4 | Terrain Classification

The importance of terrain classification in planetary missions has become increasingly apparent—most notably demonstrated by NASA's MER Spirit, which ended its mission in 2009 after becoming mired in soft soil. Unlike conventional object detection tasks, vision-based terrain classification must address subtle variations in surface features and the often-similar color profiles of different terrain types, making older classification methods less reliable for planetary rovers. Consequently, robust terrain-classification systems have emerged as a critical complement to hazard avoidance, aiming to prevent immobilization events and enhance overall mission safety.

CV techniques—such as object detection or semantic segmentation (see Section 3)—provide the means to label bounding boxes or individual pixels according to terrain classes.

Complementary approaches using proprioceptive sensors (e.g., wheel slip measurements and vibration data) can further augment classification accuracy, particularly in low-visibility or high-dust environments. A comprehensive overview of terrain assessment methods, including proprioceptive-based techniques, is presented in Sancho-Pradel and Gao (2010). More recently, Nampoothiri et al. (2021) offer an expanded survey of classification strategies that integrate both visual and nonvisual sensing (e.g., vibration and LiDAR), highlighting the growing trend toward multimodal solutions. Autonomous terrain classification and hazard detection are key enablers of robotic sampling and in situ science. Recent reviews, such as T. Zhang et al. (2019), summarize the rapid progress in extraterrestrial regolith-sampling robots and emphasize how perception and control integration have become critical for autonomous sampling on the Moon, Mars, and asteroids.

2.3.4.1 | Recent Advancements

- *Semantic segmentation and object detection:* AI-driven algorithms (e.g., Fully Convolutional Neural Network [FCNN], U-Net, and DeepLab) have begun to automate terrain classification, hazard detection, and resource identification, enhancing rover situational awareness (Swan 2021; Barrett et al. 2022; Ono 2016; Palafox et al. 2017; Rothrock et al. 2016; DeLatte et al. 2019; Tewari et al. 2022; Silburt et al. 2019; Pohlen et al. 2016; M. Yang et al. 2018; J. Long et al. 2015; Zhou et al. 2020; Goh et al. 2022; Song and Shan 2008; Ono et al. 2015; Furlán et al. 2019; Kuang et al. 2021).
- *Hybrid localization methods:* Combining MCL, deep learning, and VO can align onboard imagery with orbital maps, achieving more accurate global and local state estimates (Van Pham et al. 2003; Geromichalos et al. 2020; Dinsdale et al. 2022; Sheshadri et al. 2012; Hwangbo et al. 2009; Boukas et al. 2018; C.-S. Chen et al. 1999; Naguib 2018; Chung et al. 2018a; Wu 2019; Franchi and Ntagiou 2021; Matthies et al. 2022).

2.3.4.2 | Persistent Limitations

- *Limited data set availability:* The scarcity of labeled planetary images hinders broad generalization and robust training of advanced ML or deep-learning models (Swan 2021; Song and Shan 2008).
- *Real-time computational performance:* Onboard resources necessitate significant optimizations to run vision pipelines at operational frequencies (Matthies et al. 2022).
- *Generalizability and robustness:* Algorithms may fail to adapt if surface conditions—dust accumulation, lighting extremes, or unexpected geological features—differ significantly from training data (Boukas et al. 2018).

Table 3 presents quantitative evidence of terrain-classification challenges in planetary exploration. It shows how limited labeled data restrict model scope, onboard hardware slows real-time inference, and cross-data set tests reveal poor generalization under new lighting or terrain conditions. These metrics

TABLE 3 | (Terrain-classification) quantitative illustration of persistent limitations.

Limitation	Representative method/data set	Accuracy (%)	Inference time (FPS)	Platform/hardware	Observation/constraint	Cited paper(s)
Limited data set availability	SegMarsViT—vision-transformer classifier on Mars terrain images	94.66% on MSLNet data set	—	GPU (RTX 2080 Ti in training)	Data set limited to ≈ 6000 Mars images $\rightarrow$ weak generalization	Lv et al. (2022)
Real-time computational performance	L-ViTeX—lightweight visual-intuition transformer	99.2% overall terrain accuracy	≈ 180 ms/frame (~ 5.5 FPS)	Embedded GPU (≤ 15 W SoC board)	Meets near real-time rate but lags 30 Hz target	Mazumder et al. (2024)
Generalizability and robustness	Domain-adaptive CNN trained across HiRISE and Perseverance data sets	$\approx 96\%$ ($\pm 1.2\%$) mean IoU across domains	≈ 50 ms/frame (20 FPS)	Jetson AGX Orin (30 W)	Performance drops $> 10\%$ under unseen illumination/terrain textures	Olivi et al. (2025)

Abbreviations: CNN, Convolutional Neural Network; FPS, frames per second; GPU, Graphics Processing Unit; IoU, Intersection over Union; MSLNet, Mars Science Laboratory Network; SoC, System-on-a-Chip.

underline the need for larger data sets and lightweight adaptive models.

2.3.4.3 | Future Research Directions

- *Advanced synthetic data:* Improving the realism and diversity of simulated terrains to augment limited real data and facilitate robust training (Naguib 2018).
- *Lightweight neural networks:* Streamlining deep-learning models for low-latency execution on radiation-hardened processors (Goh et al. 2022).
- *Integrated sensor fusion:* Combining visual, proprioceptive, and multispectral sensors (e.g., thermal, LiDAR) to bolster reliability across varied surface environments (Franchi and Ntgiou 2021).
- *Adaptive, iterative updates:* Enabling onboard retraining or online learning to dynamically improve accuracy as rovers encounter new terrain types (Matthies et al. 2022).

By leveraging these localized approaches (VO and DEM-based mapping) and ongoing deep-learning innovations, planetary rovers can achieve higher autonomy and more reliable performance in challenging surface conditions.

3 | AI-Enabled CV Techniques for Space Missions

3.1 | Image Classification

Image classification is a fundamental CV task, assigning labels to entire images based on their overall content. In space robotic missions, pure classification approaches alone have limited applicability because, in many scenarios, systems require not only class labels but also the precise spatial location of objects of interest (e.g., craters and rocks). Consequently, image classification tasks in space often evolve into more detailed object detection or segmentation solutions.

3.1.1 | Advances Addressing Space Mission Requirements

- *Content-based scientific image classification:* NASA's Planetary Data System (PDS) Imaging Atlas utilizes frameworks like AlexNet for organizing and retrieving Mars imagery. Transfer learning is used to adapt Earth-based models to Martian data. Although effective for orbital imagery, classifying surface images remains challenging due to greater scene variability and a requirement for periodic retraining (Krizhevsky et al. 2017; NASA PDS Imaging Atlas; Wagstaff et al. 2018). Recent work has extended these efforts by exploring more advanced architectures and hardware-aware deployment strategies. Kharsa et al. (2023) presented a survey on quantum machine learning for image classification, highlighting how quantum-enhanced deep learning can accelerate high-dimensional image recognition for space-based applications where computational resources are limited. Their study demonstrates how quantum parallelization could support onboard processing of large imagery data sets in future missions.
- Similarly, Chase et al. (2024) profiled 18 vision-based deep-learning architectures on NASA's radiation-hardened

SpaceCube platforms. By evaluating CNN and Transformer models in terms of power usage, memory footprint, and execution speed, they demonstrated the trade-offs required for real-time terrain classification and hazard detection in orbit and on the surface. Kale et al. (2022) conducted a comparative study on Mars imagery classification using EfficientNet, Vision Transformers, and dual-attention networks trained on NASA Curiosity rover data. Their results show that knowledge transfer from Earth-based data sets significantly boosts classification accuracy on Martian terrain, making the approach promising for autonomous hazard detection and navigation. Goh et al. (2023) introduced a self-supervised distillation framework that combines representation learning and model compression to produce lightweight classifiers capable of running onboard planetary robots. Their approach reduces the need for large annotated data sets while maintaining high accuracy for terrain-classification and segmentation tasks.

- *Applicability to 3D tasks:* For certain space applications (e.g., identifying specific targets during missions), standard 2D image classification is replaced by 3D object detection or pose estimation methods. For example, Chekakta et al. (2022) demonstrated precise landing through LiDAR-based 3D pose estimation, while Piazza et al. (2022) used monocular images and deep learning for uncooperative spacecraft navigation—highlighting the essential role of 3D perception in modern space robotics. These techniques, which incorporate both classification and localization, are often more pertinent to autonomous navigation and scientific objectives. In contrast to 2D classification, 3D approaches provide the precise geometry and orientation of objects in the environment, enabling:

1. Accurate hazard detection and avoidance during rover traversals or landings (e.g., identifying rocks or craters in 3D space).
2. Robotic manipulation and sample collection, where knowing an object's pose is essential for grasping or drilling.
3. Enhanced terrain mapping, as 3D data, allows for more robust and detailed environmental modeling, crucial for path planning.

3.1.2 | Persistent Challenges and Limitations

- *Inadequate generalizability:* Models trained on Earth-based imagery (e.g., ImageNet) often struggle when transferred directly to planetary domains with significant environmental and appearance differences. Even transfer-learned architectures, such as those evaluated by Kale et al. (2022), show reduced accuracy in unseen terrain types or lighting conditions, highlighting the need for domain adaptation strategies.
- *Computational restrictions:* Deep-learning models such as AlexNet can be too large for onboard processors, limiting the rate at which classification inferences can be run in real time. Profiling results from SpaceCube tests (Chase et al. 2024) show that transformer-based models consume significantly more power than optimized CNNs, emphasizing the importance of energy-efficient architectures for future deployments.

- *Limited labeled data:* Space missions typically lack sufficient labeled data sets for robust supervised training, particularly for surface-level tasks. Self-supervised distillation techniques (Goh et al. 2023) and synthetic data generation approaches are emerging as effective methods to mitigate this bottleneck, enabling onboard learning from unlabeled imagery.

Table 4 quantifies key limitations in space-based image classification. It shows that terrestrial-trained models lose accuracy on planetary data, transformer architectures exceed energy limits on flight hardware, and self-supervised methods partially overcome scarce labeled imagery. These figures underscore the need for domain adaptation and efficient onboard learning.

3.1.3 | Future Research Directions

- *Lightweight neural architectures:* Development of more computationally efficient and size-optimized CNNs suitable for onboard environments. Recent quantum ML methods suggest that quantum-inspired lightweight models may offer significant speed and energy advantages for future deep-space hardware deployments (Kharsa et al. 2023). Supporting this direction, a parallel CNN architecture developed by Tumpa and Islam (Tumpa and Islam 2024) demonstrated over 97% classification accuracy on satellite imagery data sets while using fewer than 1 million parameters, highlighting its viability for edge deployment in satellite systems.
- *Improved synthetic data and transfer learning:* Mitigating the domain gap through advanced data augmentation and unsupervised or self-supervised learning approaches that lessen reliance on large labeled data sets. Combining transfer learning with self-supervised distillation (Goh et al. 2023) and cross-domain knowledge transfer (Kale et al. 2022) could lead to more resilient models that adapt autonomously to Martian or lunar terrain shifts.
- *Continuous onboard learning:* Implementing incremental or reinforcement learning methods that allow classification models to update themselves autonomously over the mission's lifespan, maintaining accuracy despite evolving terrain or lighting conditions. Future missions could benefit from hardware-aware profiling (Chase et al. 2024) combined with quantum-inspired optimization (Kharsa et al. 2023), allowing adaptive model updates within tight power and compute budgets.

By prioritizing flexible and lightweight architectures, leveraging transfer learning, and exploring continuous onboard updates, image classification can become more viable for targeted applications in resource-constrained space environments.

3.2 | Object Detection

Object detection extends the concept of classification by identifying and localizing multiple objects within a single image, providing bounding boxes alongside class labels. In space robotics, object detection is essential for critical functions, such as hazard detection, obstacle avoidance, and sample acquisition. Notably, it can also serve as an initial step toward pose

TABLE 4 | (Image classification): Quantitative illustration of persistent challenges and limitations.

Limitation	Representative method/data set	Accuracy (%)	Observation/constraint	Cited paper(s)
Inadequate generalizability	Semisupervised framework using only 161 training images	91.1% (vs. 81.9% baseline)	Shows effective use of unlabeled data to boost classification accuracy under scarce labeling conditions	Goh et al. (2022)
Computational restrictions	Segmentation benchmark across multiple Mars regions	Average IoU: 52.52% (in-distribution) and 42.55% (cross-region)	Indicates a strong drop in performance when moving to new regions (i.e., generalization barrier)	Purohit et al. (2024)
Limited labeled data	Lightweight transformer for Mars terrain classification (Light4Mars-T)	Highest accuracy: 89.84%	Demonstrates transformer architecture adapted for Mars terrain classification, but lacks explicit inference/power figures, and shows that achieving high accuracy remains challenging under constrained models	Xiong et al. (2024)

Abbreviation: IoU, Intersection over Union.

estimation, which is crucial for tasks ranging from orbital navigation to planetary sample retrieval.

3.2.1 | Recent Advancements

- *Evolution of deep learning for object detection:* Recent developments in DNNs, accelerated by autonomous driving research, have significantly boosted performance in space-relevant tasks. Commonly applied detection architectures include Region-Based Convolutional Neural Network (Girshick et al. 2014), You Only Look Once (YOLO) (Redmon et al. 2016), and Single-Shot Detector (SSD) (W. Liu et al. 2016). Other less conventional methods, such as Deep Universal Graph embedding neural network (Verma and Zhang 2019), learn task-agnostic embeddings through adaptive supervised learning, supporting improved transfer learning. Duba et al. (2024) developed a stereo-vision-based SSD system that performs real-time 3D obstacle detection and triangulation for autonomous spacecraft navigation. It delivers depth estimation in free space, crucial for debris removal and robotic path-planning applications.

De Luca Visioli and De Figueiredo Pereira Alves Taveira Pazelli (2024) introduced a lightweight OpenCV-based monocular system for CubeSat detection and tracking. Using background subtraction and edge-geometry analysis, their method enables robotic manipulators to visually interact with nanosatellites. Grauer et al. (2024) proposed an approach for detecting uncooperative space debris and inactive satellites by combining YOLOv8 with Fast-SCNN and visual-foundation-model features. Their system proved robust under variable lighting and long-range imaging conditions. Gizzi et al. (2025) highlighted NASA's On-Board Artificial Intelligence Research (OnAIR) framework, which integrates detection and recognition models directly into spacecraft avionics. OnAIR streamlines onboard AI deployment and enables autonomous detection tasks on operational missions.

- *Crater detection for navigation:* One major application involves crater identification, a vital component of lunar or Martian absolute navigation. Approaches like the one-stage anchor-free Dense Point Crater Detection Network (Zihao and Jiang 2021) and CraterIDNet (Wang et al. 2018) have displayed strong detection accuracy across multiple crater scales. Alternatively, crater segmentation solutions leveraging a binary mask (e.g., modified U-Net) focus on crater edges, assisting subsequent matching or pose estimation (Downes et al. 2020; Silburt et al. 2019, 2018; Benedix et al. 2018; Cohen et al. 2016; PyCDA; L. Singh and Lim 2008). These methods offer F1 scores exceeding 90%, indicating reliable detection performance.
- *Sample tube detection in MSR:* A key two-stage approach addresses the identification and pose estimation of sample tubes for ESA's MSR. A lightweight YOLO variant (YOLOv3-tiny) localizes tubes in RGB images, after which classical CV techniques (e.g., Sobel filtering, adaptive thresholding, and Perspective-n-Point [PnP] with RANdom SAMple Consensus [RANSAC]) refine pose estimates (Papon et al. 2017; Daftry et al. 2021; Alvarez et al. 2022; Mantoani et al. 2022). Similar work evaluated template

matching and CNN-based segmentation, finding that semantic segmentation often outperforms handcrafted features in challenging Martian environments.

- *3D rock detection for hazard analysis:* Object detection also extends to 3D tasks, such as identifying rocks and obstacles in planetary terrain. Specialized SSD-based architectures have been proposed to detect rocks in stereo or RGB-D imagery, with modifications to the backbone networks (e.g., VGG-16 and ResNet-50) for speed and accuracy (W. Liu et al. 2016; Furlán et al. 2020; Gao et al. 2013). Successful 3D detections inform hazard avoidance, enabling safer, more efficient rover path planning. In Furgale et al. (2012), the authors employ a number of modified SSD network architectures to address the problem of rock detection as a 3D object detection problem rather than one of segmentation (see Figure 2).

3.2.2 | Key Unresolved Challenges

- *Robustness and adaptability:* Despite successes, detection models can degrade with changing lighting, dust, and partial occlusions—common in extraterrestrial terrains. Grauer et al. (2024) emphasize that detecting inactive spacecraft and debris involves severe domain shifts due to illumination and background variability. C. Liu et al. (2024) introduce a domain adaptation framework that uses confidence-driven region mixing and adversarial learning to improve object detection robustness under changing environmental conditions, such as lighting and background noise—closely mirroring the domain shifts encountered in planetary missions.
- *Computational efficiency:* Most deep detection networks are resource-intensive, necessitating specialized optimizations or lightweight architectures to achieve real-time inference on flight processors. Results from De Luca Visioli and De Figueiredo Pereira Alves Taveira Pazelli (2024) and Gizzi et al. (2025) demonstrate that optimized OpenCV pipelines and onboard AI frameworks can achieve real-time inference under power constraints.
- *Data set scarcity and domain shifts:* Labeled data remains limited, particularly for planetary environments. Models trained on Earth-based sets or simulations often struggle when confronted with novel terrain variations. Hybrid simulation and foundation models (Grauer et al. 2024) offer promising directions for data diversification and adaptation. Duraklı and Aptoula (2023) propose a domain generalization approach that improves object detection performance on unseen remote-sensing environments without needing target domain labels—highlighting potential for space missions facing novel terrain and limited annotated data.

Table 5 compares recent object detection methods for planetary imagery, showing that YOLO-based models are the most practical for real-time crater detection on embedded systems, like, Jetson and Edge TPU, while EfficientDet achieves strong accuracy for space-object recognition in simulated environments. Meanwhile, Transformer-based detectors deliver the highest precision and recall on lunar and Martian data sets, outperforming YOLO in accuracy but requiring larger Graphics Processing Unit (GPU) resources, limiting their onboard applicability.

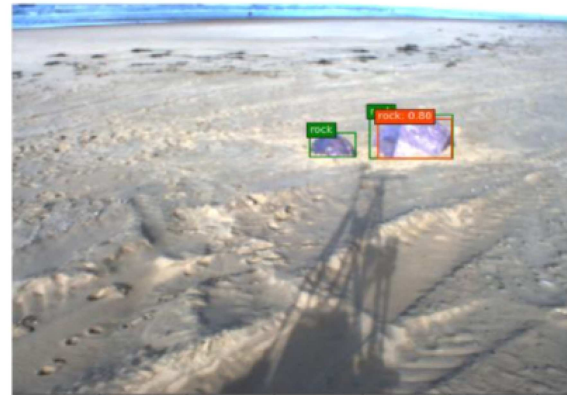
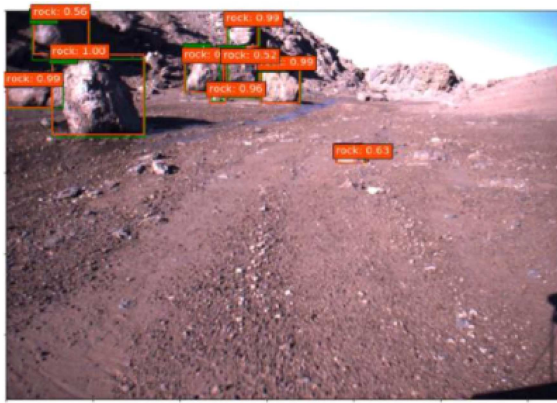


FIGURE 2 | Rock detection performed (left, Devon Island [Furgale et al. 2012]; right, Katwijk beach [Hewitt et al. 2018]).

3.2.3 | Future Research Directions

1. *Flight-friendly architectures*: Further efforts are needed to prune, quantize, or compress detection networks (e.g., YOLO and SSD) for onboard deployment. Integration of stereo SSD methods (Duba et al. 2024) and quantized foundation models (Gizzi et al. 2025) could balance accuracy and efficiency.
2. *Hybrid pipelines*: Combining learning-based detection with classical geometric techniques (e.g., PnP and RANSAC) can mitigate domain shifts and sensor artifacts. Harvard et al. (2020) proposed a hybrid architecture using CNN-trained keypoints with PnP-RANSAC refinement, improving stability under difficult visual conditions.
3. *Synthetic data and transfer learning*: Strategies like domain randomization, unsupervised adaptation, and few-shot learning help address data scarcity. Leveraging visual-foundation models for few-shot adaptation can enhance mission robustness. For example, C. Liu et al. (2024) introduced a domain adaptation framework using confidence-driven region mixing and adversarial learning to improve object detection robustness under changing lighting and background—conditions similar to those in space missions.
4. *Adaptive detection frameworks*: Enabling incremental updates or continuous learning could maintain model relevance over the course of long-duration missions. Grauer et al. (2024) combine YOLOv8, Fast-SCNN, and visual-foundation models, showcasing a hybrid pipeline that adapts to lighting and sensor noise variations, which aligns with the goals of continuous learning in long-duration missions.

By integrating state-of-the-art deep-learning algorithms with classical vision techniques, object detection holds a pivotal role in sustaining autonomy and situational awareness across orbital and planetary missions, from crater-based navigation to sample fetching and beyond.

3.3 | Semantic Segmentation

Semantic segmentation classifies every pixel in an image according to its associated class, generating a detailed scene understanding crucial for space robotics. In planetary exploration

specifically, this capability is vital for ensuring safe navigation, identifying scientific targets, automating hazard avoidance, and facilitating tasks, such as landing site assessment.

3.3.1 | Advances and Applications

- *Image Segmentation as an Extension of Classification*: Unlike simple classification, which assigns a single label per image, segmentation outputs a pixelwise mask. This enables precise identification of geological features, such as craters or rocks, which is essential for tasks, like, Mars landing site selection (Barrett et al. 2022; Ono 2016), rover path planning (Palafox et al. 2017; Rothrock et al. 2016), and automated crater or rock labeling (DeLatte et al. 2019; Tewari et al. 2022; Silburt et al. 2019).

Qu et al. (2025) present a systematic review of Martian image segmentation methods from 2019 to 2025, categorizing approaches for rover vision and remote-sensing imagery and highlighting data set gaps. Parmar et al. (2024) propose an attention-based transformer architecture (MANet) for segmenting orbiting spacecraft under solar glare and low lighting, achieving near-perfect IoU on the SPARK2024 data set. H. Liu et al. (2025) introduce SpaceSeg, a multiattention segmentation framework for multispacecraft scenarios that reaches state-of-the-art accuracy on the SpaceES data set. Kuang et al. (2022) conducted a review of semantic terrain segmentation for planetary rovers, stressing the importance of domain adaptation for safe path planning.

- *Key Architectures (FCNN, U-Net, DeepLab, ConvDeconv, FR-RNs, and DenseASPP)*: Many networks have been adapted to planetary scenarios, including FCNNs, U-Net, DeepLab, and ConvDeconv. Research has further introduced approaches, like, Full-Resolution Residual Networks (FR-RNs) (Pohlen et al. 2016) and DenseASPP (M. Yang et al. 2018), which incorporate atrous convolution and multiscale pooling to handle large scenes with varying features. These advanced models demonstrate a rapidly evolving state of the art.
- *Crater Detection and Landing Site Selection*: Crater segmentation is integral to absolute navigation and hazard detection for missions targeting the Moon or Mars (Downes et al. 2020; Silburt et al. 2019, 2018; Benedix et al. 2018;

TABLE 5 | Quantitative comparison of YOLOv3-tiny and classical approaches.

Model/method	Task/data set	mAP/accuracy	Platform/hardware	Key finding	Cited paper(s)
YOLO-series (YOLOv5/YOLOv8)	Evaluation of resource-efficient crater detectors	0.889%–0.935%	Google Coral Edge TPU, AMD Versal SoC VCK190	Demonstrates the viability of YOLO models for Mars crater detection in embedded hardware	Vellas et al. (2024)
EfficientDet (EfficientNet backbone)	Space-object detection (satellites/debris) with SPARK data set	Detection: mean precision 78.45%, mean recall 92.00%	Uses EfficientDet+ EfficientNet v2 in simulation	Shows that EfficientDet-based method gives strong detection/recognition of space objects under challenging backgrounds	AlDahoul et al. (2022)
RT-DETR (Transformer-based detector)	Rockfall detection on Moon and Mars (RMaM-2020 data set)	mAP50 = 0.964, precision = 0.982, recall = 0.955	—	Outperforms YOLO-series models by > 40% in relative performance on small targets in low-contrast planetary imagery	Zang et al. (2025)

Abbreviations: mAP, mean average precision; SoC, System-on-a-Chip; TPU, thermoplastic polyurethane; YOLO, You Only Look Once.

Cohen et al. 2016; PyCDA; L. Singh and Lim 2008). Methods such as DeepMoon and LunaNet help automate the traditionally manual process of mapping craters, yielding robust landmark identification for safe EDL operations.

- *Rock and Hazard Segmentation:* Segmenting rocks and obstacles significantly supports geological analysis and hazard avoidance on planetary surfaces (Song and Shan 2008; Ono et al. 2015; Furlán et al. 2019; Kuang et al. 2021). For instance, a two-phase segmentation approach (texture classification followed by active contours, Song and Shan 2008) refines rock boundaries, although partial coverage by soil or overlapping rocks poses continued challenges. In some cases, real-time frameworks (e.g., modified U-Net variants) can deliver subsecond inference, a critical factor for onboard autonomy (Furlán et al. 2019). Similar approaches are employed for extended slip prediction (soil property and object classification [SPOC], Rothrock et al. 2016) and morphological analysis (e.g., ephemeral rock shapes) on rovers.

In Song (2008), rock segmentation is performed in two phases: texture-based image segmentation and active contours-based boundary refinement. After texture feature classification, clustering is performed to put rough outlines on rocks, which are then further refined by performing edge-flow boundary detection on each identified rock, allowing them to be cleaned up with active contours. This is an interesting approach to the problem, but it struggles against rocks that might have soil partly covering rocks or rocks that are behind each other, for example, a large rock is half-identified due to the boundary edge detection problem, as demonstrated in the third picture in Figure 3.

- *Application to Slip Prediction and Scene Understanding:* Beyond static terrain labeling, segmentation-based methods such as SPOC and its variants (SPOC-G and SPOC-R) (Rothrock et al. 2016) incorporate slip prediction, wheel-track detection, and scale considerations for better rover autonomy. Contrastingly, multinet network ensembles (e.g., combining various ResNet-50 modifications, Zhou et al. 2020) have been shown to improve classification accuracy further.
- *Limited Data set Solutions (Synthetic Data and Transfer Learning):* Activities like AI4Mars (Swan 2021) show how crowdsourced labeling and advanced architectures (DeepLabv3+ with a ResNet backend) can yield strong segmentation results, even with relatively small class sets. Meanwhile, specialized contrastive pretraining with ResNets has demonstrated effectiveness in bridging domain gaps and mitigating overfitting (Goh et al. 2022).

3.3.2 | Persistent Limitations

1. *Computational Complexity:* Most deep segmentation algorithms (e.g., FR-RNs and DenseASPP) require substantial computational resources, exceeding typical flight hardware capabilities. Techniques such as multiresolution image processing and model compression are often necessary. SpaceSeg (H. Liu et al. 2025) demonstrates high segmentation performance but relies on multiattention

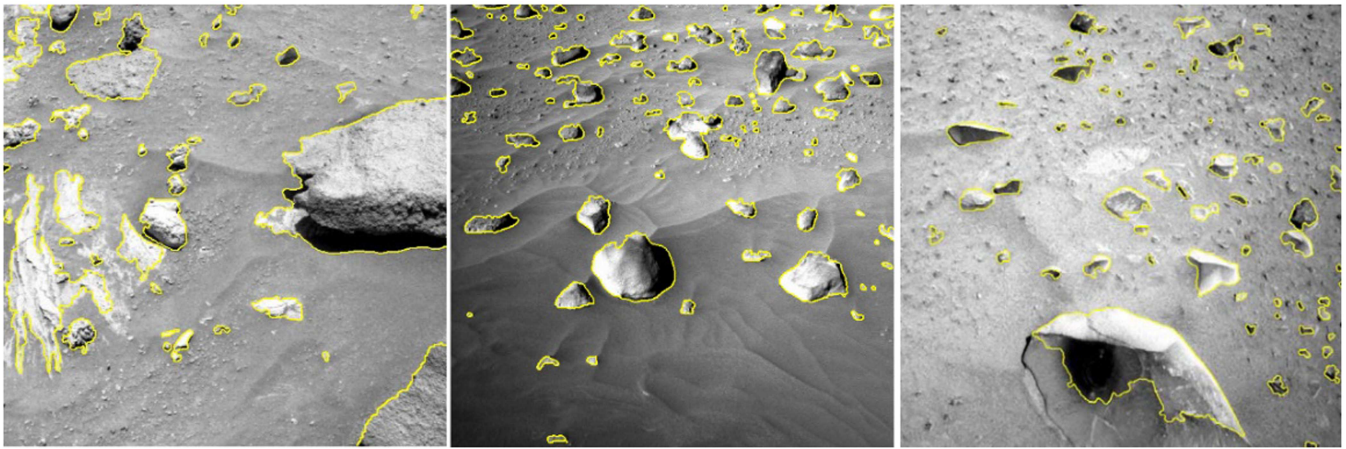


FIGURE 3 | Rock classification performed in Song (2008).

modules and deep transformer-based encoders, which significantly increase computational demands, directly reflecting the limitation that advanced architectures like FR-RNs and DenseASPP exceed typical flight hardware capabilities.

2. *Data Scarcity and Diversity*: Adequate labeled planetary imagery for supervised training is limited, particularly for specialized tasks like rock segmentation or crater boundary detection. Models trained on Earth-based or purely synthetic data face domain shifts on real planetary terrains. Qu et al. (2025) emphasize that data set heterogeneity remains the main limiting factor for cross-mission generalization.
3. *Overfitting and Adaptability*: Models can overfit to specific terrain types or conditions (lighting and dust coverage). Without domain adaptation or advanced augmentation, they may fail in unseen environments. Kuang et al. (2022) highlight that segmentation models trained on limited or single-domain planetary data sets tend to overfit to specific terrains (lighting and dust conditions), underscoring the need for domain adaptation and augmentation to improve generalization—precisely aligning with point 3 on overfitting and adaptability.

Table 6 quantifies the main limitations of semantic segmentation across diverse domains. It shows that even efficient models like RepMPSeg achieve real-time performance, but computational trade-offs remain critical for high-resolution tasks. Data scarcity continues to hinder generalization across domains, though methods like MetaPix improve transferability by weighting synthetic data effectively. Meanwhile, advanced models such as BLO-SAM address overfitting under limited data conditions using bi-level optimization. These findings emphasize that despite progress, semantic segmentation still involves a core trade-off between accuracy, generalizability, and real-world deployability.

3.3.3 | Future Research Directions

1. *Lightweight, Efficient Models*: Continued work on pruning, quantization, or specialized architectures tailored to radiation-hardened onboard CPUs/FPGA-based accelerators for real-time inference (Goh et al. 2022; H. Liu et al. 2025).

2. *Enhanced Synthetic Data and Augmentation*: Employing Generative Adversarial Network (GAN)-based approaches, domain randomization, and advanced data augmentation strategies for better coverage of varying terrain conditions, mitigating overfitting (Goh et al. 2022; Naguib 2018).
3. *Adaptive Segmentation*: Development of real-time segmenters capable of integrating multisensor data (e.g., thermal, LiDAR) and adjusting to changing conditions, such as lighting shifts or dust storms. Adaptive multi-attention fusion as demonstrated in *SpaceSeg* (H. Liu et al. 2025) and *MANet* (Parmar et al. 2024) to maintain accuracy under illumination changes.
4. *Continuous Onboard Learning*: Incremental model updates using newly acquired data and offline or partially onboard retraining, reducing domain mismatch over long-duration missions (Matthies et al. 2022; Kuang et al. 2022).

By combining robust, lightweight semantic segmentation models with advanced data augmentation and adaptive learning techniques, planetary explorers can achieve more reliable hazard detection, better scientific returns, and safer landings in harsh, evolving extraterrestrial terrains.

3.4 | Relative Pose Estimation

Relative pose estimation determines an object's position and orientation relative to a sensor or another object, playing a critical role in tasks, such as autonomous spacecraft rendezvous, docking, small-body proximity operations, and rover manipulation. In planetary scenarios, it often augments or follows object detection to obtain precise 3D transformations, while in orbital contexts, it underpins safe approach and capture maneuvers.

3.4.1 | Advances and Applications

- *Relationship to 3D CV Methods*: Relative pose estimation is central to many 3D CV tasks (Structure-from-Motion, Agarwal 2011; Schonberger and Frahm 2016; simultaneous localization and mapping [SLAM], Durrant-Whyte and Bailey 2006; VO, Nistér et al. 2004). Traditional pose

TABLE 6 | (Semantic segmentation): Quantitative illustration of persistent limitations.

Limitation	Representative method/data set	IoU/accuracy (%)	Inference time/latency	Observation/constraint	Cited paper(s)
Computational complexity	RepMPSeg—Real-time multipath segmentation on Cityscapes and CamVid	77.4 (Cityscapes), 78.6 (CamVid)	91.4 FPS (Cityscapes), 117.33 FPS (CamVid)	Achieves real-time performance with optimized feature fusion for high and low-resolution branches	Pengfei et al. (2025)
Data scarcity and diversity	MetaPix—Meta pixel weighting of synthetic + real data for improved domain transfer	Improves mIoU over synthetic-only baseline	—	Learns optimal pixel-level weighting to bridge the domain gap between synthetic and real images	Jian and Gao (2021)
Overfitting and adaptability	BLO-SAM—Bi-level optimization finetuning of Segment Anything Model for downstream medical segmentation	Outperforms baseline SAM finetuning (exact IoU not stated)	—	Reduces overfitting by separating prompt embedding and model weights training using dual subsets	L. Zhang, Liang, et al. (2024)

Abbreviations: FPS, frames per second; IoU, Intersection over Union; mIoU, mean Intersection over Union.

estimation often relies on local keypoints and descriptors, but modern approaches increasingly employ DNNs.

At its core, relative pose estimation can be framed as solving for the transformation (rotation and translation) that aligns corresponding features between two views or between a model and a sensor image. Classical solutions involve epipolar geometry, the essential/fundamental matrix, and PnP formulations. More recently, end-to-end deep networks predict 6D poses directly or refine initial geometric estimates with learned priors (Kehl et al. 2017; Xiang et al. 2018). Such methods improve robustness when keypoints are sparse or environmental conditions (e.g., lighting and occlusion) degrade traditional descriptor matching.

- Deep-Learning Approaches:** ESA's Spacecraft Pose Estimation Challenge showcased multiple CNN-based solutions, including the SPN (Sharma and D'Amico 2019), pose estimation with deep landmark regression (B. Chen et al. 2019), soft-classification (Castano et al. 2003), and segmentation-driven approaches (Gerard 2019). SPN, for instance, uses classification and regression subnetworks to predict bounding box coordinates and discretized rotations, while subsequent components estimate translation from bounding box dimensions. Keypoint-based methods (e.g., B. Chen et al. 2019) regress 2D features for geometry-based alignment, whereas segmentation-driven approaches (Gerard 2019) combine region segmentation with iterative PnP and RANSAC. Bechini and Lavagna (2025) proposed a single-CNN multitask architecture performing ROI detection and keypoint regression for monocular pose estimation with exceptional efficiency. Piazza et al. (2022) introduced a deep pipeline combining object detection and keypoint geometry for uncooperative target pose estimation using Spacecraft Pose Estimation Dataset (SPEED) data, achieving centimeter-level precision.

Beyond these specialized spacecraft-focused methods, researchers have introduced general-purpose 6D pose architectures (Kehl et al. 2017; Xiang et al. 2018; K. Park et al. 2019) that utilize pixelwise coordinate regression or feature learning to infer object orientation and position in a single pass. These frameworks often leverage large synthetic training sets or domain randomization to handle limited real-world data (Tobin et al. 2017). The advantage is a more robust and data-driven pipeline that learns to correct common errors (e.g., partial occlusion) by incorporating 3D-aware features. I. P. Singh et al. (2025) developed a Supervised Domain Adaptation framework for 6-DoF pose estimation, bridging the synthetic-to-real gap through domain-invariant feature learning and risk optimization. C. Long and Hu (2022) presented a multicircular-feature monocular pose method for noncooperative spacecraft, using ellipse geometry to achieve accurate 6-DoF orientation.

Harvard et al. (2020) proposed an architecture that uses existing keypoint specialized algorithms to identify robust keypoints and then train a CNN on this limited set of keypoints (with feature descriptor components) to create specialized descriptors. For each landmark, a visibility map is also generated through ray tracing. PnP-RANSAC was used to estimate the pose and nonlinear filter to track the

pose. A comparative analysis of keypoint-based and nonkeypoint-based deep-learning approaches for spacecraft pose estimation was presented in Rathinam and Gao (2020), see Figure 4.

- **Hybrid Pipeline Examples:** Many pipelines combine learning-based detection with classical CV modules (PnP, ICP, and RANSAC) to finalize pose estimates. For instance, using an object detection CNN to localize a satellite, then cropping and applying keypoint regression or geometry-based alignment steps. Typical hybrid designs reduce end-to-end complexity while retaining the strengths of geometry-based refinement. A network might yield an approximate pose or keypoint set, which is then iteratively refined via RANSAC or iterative closest point (ICP) methods (Xiang et al. 2018). This approach balances the speed of CNN inferences with the accuracy of classical optimization, ensuring a stable final pose even if the initial prediction is imperfect.

Importantly, several hybrid pipelines have been demonstrated in actual space mission contexts. For example,

NASA's Seeker-1 CubeSat used a real-time hybrid approach to estimate the relative pose of the Cygnus spacecraft by combining CNN-based detection with geometry-based refinements (Dhamani et al. 2020). Similarly, systems developed for ESA's SPEED Challenge utilized hybrid pipelines incorporating CNNs and geometric modules such as PnP and RANSAC to estimate the 6-DoF pose of uncooperative targets, achieving high accuracy with limited onboard resources (Piazza et al. 2022; Pauly et al. 2023).

Beyond flight-tested systems, hybrid architectures remain a key area of ongoing research. For example, Pauly et al. (2023) surveyed a wide range of hybrid designs that consistently outperform purely learning-based or classical approaches in robustness and efficiency. Likewise, Garcia et al. (2021) proposed LSPnet, a network that blends CNN-based object localization with geometric constraints to estimate 2D pose in scenarios relevant to uncooperative spacecraft operations. These examples collectively reinforce the value of hybrid pipelines as both a proven tool in real missions and a promising direction for future space robotics systems.

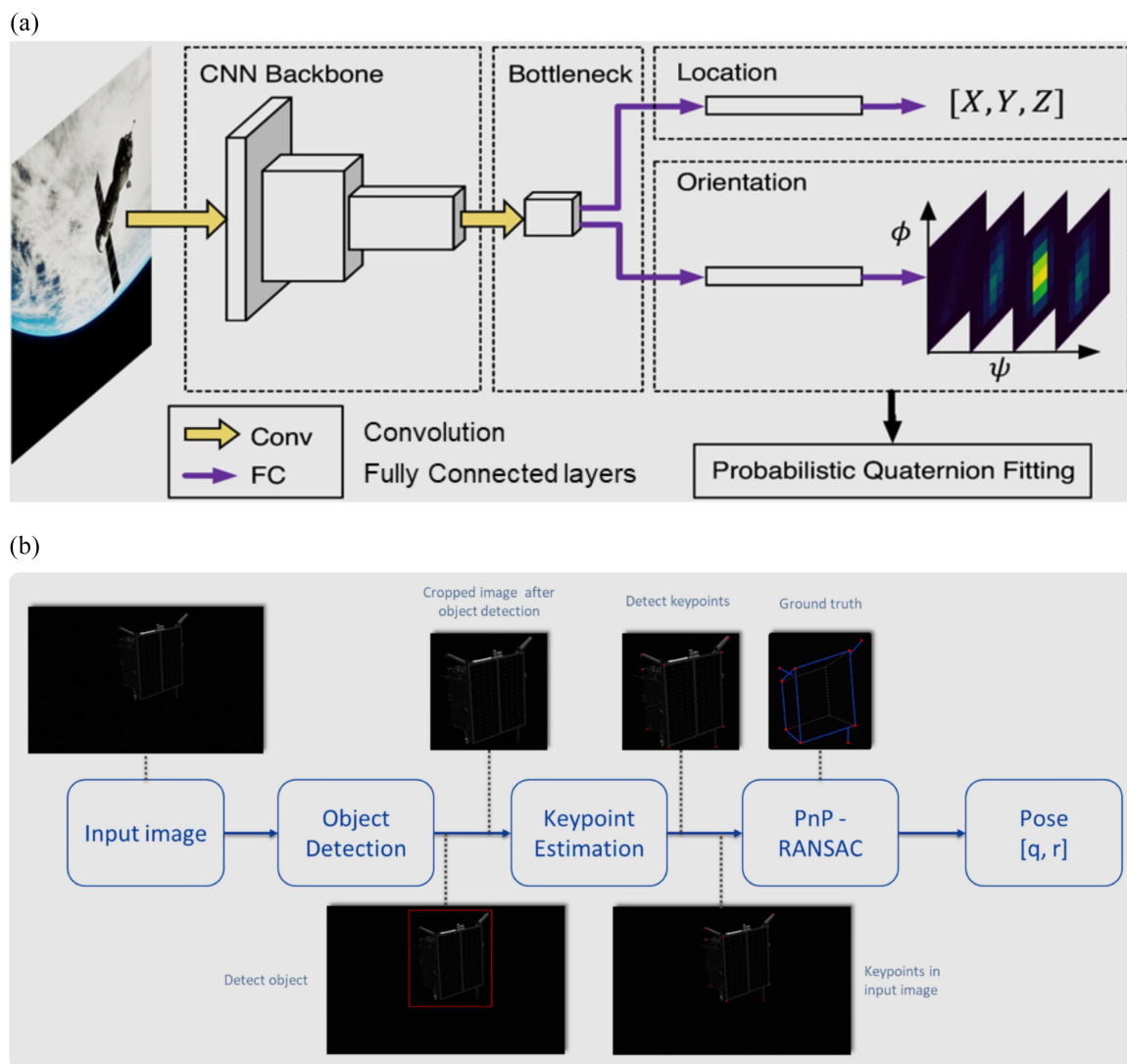


FIGURE 4 | Deep-Learning Framework using (a) nonkeypoint (Proenca and Gao 2020) or (b) keypoint method (Rathinam and Gao 2020). CNN, Convolutional Neural Network; RANSAC, RANDOM SAmple Consensus.

- *CubeSats and Low-Power Missions*: Real-time relative bearing or pose estimation under constrained hardware has been demonstrated by missions like Seeker-1 (Dhamani et al. 2020), providing bearing estimates of uncooperative targets (Cygnus) at ranges of 5–40 m and > 1 Hz. Work by Proenca and Gao (2020) and Harvard et al. (2020) further illustrates hybrid approaches combining classification, regression, and keypoint-based features, as well as specialized descriptors.

In these resource-limited scenarios, lightweight neural networks and efficient descriptor extraction are crucial. Approaches may employ network pruning, quantization, or specialized hardware accelerators to enable near real-time performance (K. Park et al. 2019). In addition, robustness to dynamic lighting and reflectance—more pronounced in space—often necessitates multisensor fusion (e.g., combining RGB cameras with infrared or LiDAR) to fill in gaps where a single sensor might fail.

3.4.2 | Persistent Limitations

- *Susceptibility to Occlusion and Lighting Variability*: Partial occlusion, reflective spacecraft surfaces, and extreme illumination differences remain challenging. Keypoint-based methods risk large errors if 2D feature detection or segmentation is inaccurate (Proenca and Gao 2020; Harvard et al. 2020). The need for precise vision-based localization under harsh lighting is also evident in regolith-sampling robots, where manipulation accuracy depends on robust visual perception (T. Zhang et al. 2019).
- *Onboard Computational Constraints*: High-fidelity models can surpass flight processor capabilities, necessitating extensive optimization (Pauly et al. 2023).
- *Generalization Challenges*: Approaches trained on particular spacecraft or scenario assumptions may struggle with new shapes, unknown small bodies, or planetary rover tasks (Wu 2019).

Table 7 quantifies key limitations in relative pose estimation systems for spaceborne applications. It shows that CNN-based pipelines can achieve high accuracy using synthetic data sets like SPEED, but still face degradation under glare and occlusion. Transformer-based models such as PVSPE offer improved robustness and accuracy, but their onboard deployment is limited by computational demands. Meanwhile, model-free estimators like DenseNet show promising generalization across satellite types, though pose accuracy can vary with unseen geometry. These findings reinforce the ongoing challenge of balancing accuracy, efficiency, and adaptability in autonomous spacecraft navigation.

3.4.3 | Future Research Directions

1. *Lightweight, Efficient Pose Estimation*: Developing quantized, pruned, or hardware-accelerated solutions for real-time inference on resource-limited spacecraft hardware (Proenca and Gao 2020; Pauly et al. 2023).
2. *Robustness and Adaptability*: Integrating synthetic or domain-randomized data sets, multisensor fusion (vision + LiDAR + IMUs), or soft-classification techniques to mitigate occlusion/lighting issues (Harvard et al. 2020).

TABLE 7 | (Relative pose estimation): Quantitative illustration of persistent limitations.

Limitation	Representative method/ data set	Pose error (translation/ rotational)	Platform/hardware	Observation/constraint	Cited paper(s)
Susceptibility to occlusion and lighting variability	Monocular CNN + keypoint regression + PnP on SPEED data set	$\sim 2.5\text{ cm}/\sim 1^\circ$	—	Performs well in synthetic-to-real, but some drop in performance under glare and partial occlusion conditions	Piazza et al. (2021)
Onboard computational constraints	Pyramid Transformer Network (PVSPE) on SPEED+ and SHIRT data sets	cm-level/deg-level	spacecraft CPU/GPU	High accuracy but complex model; not optimized for low-power real-time operation onboard spacecraft	H. Yang et al. (2024)
Generalization challenges	Satellite-model-free DenseNet+ EKF across SPEED, SHIRT, and URSO data sets	Generalizes across targets with comparable accuracy	Monocular camera + adaptive EKF	A single model trained on one satellite type performs well on unseen data sets, but performance still varies depending on geometry	Shukla et al. (2024)

Abbreviations: CNN, Convolutional Neural Network; CPU, Central Processing Unit; EKF, extended Kalman filter; GPU, Graphics Processing Unit; PnP, Perspective-n-Point; PVSPE, Pyramid Vision Spacecraft Pose Estimation; SHIRT, Satellite Hardware-In-the-loop Rendezvous Trajectory; SPEED, Spacecraft Pose Estimation Dataset; URSO, Unreal Rendered Spacecrafts On-Orbit.

3. *Generalizable Architectures*: Designing frameworks that can handle a wide variety of spacecraft geometries, uncertain small-body terrains, and rover-based pose tasks through domain adaptation or multitask learning (Chung et al. 2018a).
4. *In-Flight Updating*: Allowing partial retraining or online adaptation to incorporate newly captured data could preserve accuracy throughout extended missions (Pauly et al. 2023).

By combining deep-learning-based detection, segmentation, or keypoint regression with classical geometry refinements, relative pose estimation continues to enable robust, autonomous navigation and manipulation for both orbital and planetary applications.

3.5 | Feature Matching

The matching of traceable characteristics between locally produced maps from rover images and orbital references is a long-established approach for global localization in planetary missions. Several methods exist: direct matching of terrain topography, searching for specific, salient features (e.g., large rocks), or leveraging deep learning for map-based matching. This section follows the original structure, highlighting these three areas.

3.5.1 | Terrain Feature Matching

Terrain matching focuses on aligning local terrain-relief characteristics with global orbital data:

- *DEM-based alignment*: Van Pham et al. (2003) used a discretized particle filter and zero-mean normalized cross-correlation to match fixed-size local DEM patches (30 m × 30 m at 1 m/pixel resolution) with a global DEM, updated at 1 Hz from onboard stereo. Position uncertainty was modeled by combining VO data with a Gaussian distribution, and the approach achieved ~2 m accuracy within a 1-m/pixel orbital DEM. Similarly, GA-SLAM by Geromichalos et al. (2020) integrated global updates into an SLAM filter by down-sampling local DEM data, performing pixel-by-pixel cross-correlation with the global map (including rotational increments), yielding localization accuracies better than the DEM resolution (~0.5 m). Gnam et al. (2022) proposed an efficient feature matching and mapping method for TRN using hypothesis gating. By leveraging estimated feature positions from navigation filters, the algorithm narrows the search space, achieving an 87% reduction in matching time and improved accuracy for classical descriptors such as SIFT and ORB.
- *AGL approach*: Dinsdale et al. (2022) proposed AGL in the context of the ESA SFR. AGL compares local rover maps (DEM + orthoimage) with reference maps to reduce drift, guided by zero-mean cross-correlation and a particle filter. This system can operate at ~5 s per rover stop on LEON4 hardware, achieving up to 8 m accuracy over extended traverses. The method also accommodates multiple resolution levels for speed and robustness.

These terrain-level approaches can be computationally intensive, often requiring multiresolution strategies or particle filters to remain feasible on flight processors.

3.5.2 | Salient Feature Matching

Salient feature matching relies on aligning distinct landmarks (rocks, outcrops, brightness patterns) between orbital and local rover images:

- *Rock-based approaches*: Hwangbo et al. (2009) matched large rock features extracted from orbital imagery to local rover reconstructions. Brightness, shape cues, and a 2D affine transformation refined the rover's position estimate, functioning similarly to descent imagery localization methods used in EDL. The morphological analysis also helped reject outliers (e.g., shadows). Pan et al. (2025) expanded this line of research by integrating point and line features into monocular SLAM for spacecraft relative navigation. By using *Plücker coordinates* to represent lines robustly, the system improved accuracy by over 60% relative to ORB-SLAM and ran in real time at ~17.8 Hz.
- *Multiscale landmark matching*: A subsequent approach by Boukas et al. (2018) matched networks of salient landmarks or “blobs” across global and local maps using DARCES (C.-S. Chen et al. 1999). Hessian analysis, local entropy, and offline/online classification (*k*-Nearest Neighbors) were used to detect outcrops and large rocks. Field tests during the ESA Seeker campaign achieved ~40 m accuracy after 3 km traverses, highlighting the potential for robust large-scale localization in regions lacking craters.

Such methods are particularly important where conventional geometric structures (e.g., craters) are absent, or the terrain is dominated by rock distributions.

3.5.3 | Map-Based Feature Matching

Map-based feature matching merges classical and deep-learning strategies for robust alignment between orbital or synthetic maps and rover imagery:

- *Deep-learning expansions*: Naguib et al. (2018) proposed a conditional GAN-based approach, rendering synthetic DEM views that a CNN discriminator compares to real rover images for pose likelihood scoring, building upon the work of Van Pham et al. (2003). PLANNet by Chung et al. (2018a) and Wu (2019) uses an SNN architecture to compare rover images with candidate orbital orthoimage sections, employing a sliding-window approach. Franchi and Ntagiou (2021) further integrated an MCL filter with an SNN to avoid exhaustive image searches. These solutions demonstrate improved robustness to viewpoint and illumination changes.

Chung et al. (2018a) and Wu (2019) describe an SNN model, called PLANNet, which learns deep associations between visible salient features between reprojected rover images (four images) and corresponding orbital orthoimage regional sections, from which the rover position can be determined. Using pretrained 50-layer ResNetv2 feature extractor branches, a deep-learned feature vector is extracted from each branch and concatenated together, then fed to a fully connected layer outputting a match/no match vector, the model architecture is shown in Figure 5. A sliding-window approach is used to extract regional sections from the orbital orthoimage to feed the SNN, resulting

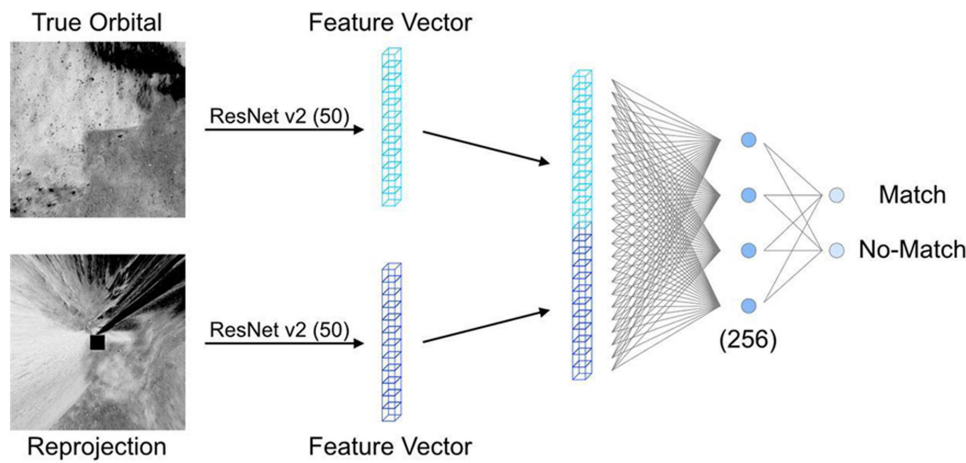


FIGURE 5 | PLanNet Siamese Neural Networks architecture (Wu 2019).

in a cumbersome, exhaustive search which does not scale well with map size. Guo et al. (2024) introduced the CtIFT deep-learning algorithm for invariant image-to-image feature matching in spacecraft docking optical imaging. It demonstrated robustness against illumination and viewpoint changes, proving highly effective for autonomous rendezvous and docking operations.

- *Hybrid crater or rock-based detection:* Some systems incorporate crater databases or rock detection to refine localization (e.g., crater detection from LiDAR/stereo data, Matthies et al. 2022) or semantic segmentation for large rocks. While these advanced methods can generalize better, they often demand significant computing resources.
- *3D key feature selection for on-orbit servicing (OOS):* Spacecraft models are simulated by comparing manual and automated 3D keypoint extraction (e.g., HarrisKeypoint3D in Point Cloud Library, they demonstrated that geometric feature matching can yield accurate pose estimation even under tight computational limits (Wessling et al. 2023).

Map-based matching leverages both classical correlation and learned representations, ensuring more adaptive performance across lighting, scale, and viewpoint variations.

3.5.4 | Advances and Applications

- *Traditional feature matching:* Techniques like SIFT, SURF, and ORB have long been mainstays in planetary navigation (Agarwal 2011; Schonberger and Frahm 2016; Durrant-Whyte and Bailey 2006; Nistér et al. 2004; Thompson and Castano 2007; Yen et al. 1999; Di et al. 2013; [Edge Detection and Image Segmentation \(EDISON\) System](#); Jin et al. 2021; Pascual et al. 2019). They remain efficient for moderate lighting changes, but can struggle with extreme domain shifts. The deep-learning-based CtIFT algorithm (Guo et al. 2024) and 3D keypoint matching frameworks (Wessling et al. 2023) extend classical approaches by embedding feature detection into neural networks and 3D geometric modeling, providing better adaptability for spacecraft docking and OOS missions.
- *Deep-learning techniques:* SNNs and GANs, along with domain-randomized training, significantly enhance

matching accuracy in harsh environmental contexts (Van Pham et al. 2003; Geromichalos et al. 2020; Dinsdale et al. 2022; Sheshadri et al. 2012; Hwangbo et al. 2009; Boukas et al. 2018; C.-S. Chen et al. 1999; Naguib 2018; Chung et al. 2018a; Wu 2019; Franchi and Ntgiou 2021; Matthies et al. 2022).

3.5.5 | Persistent Limitations

- *Environmental Variability:* Severe lighting, dust, and evolving terrain textures degrade both classical and deep-learning feature matching (Chung et al. 2018a).
- *Computational Complexity:* Deep matching models (e.g., cGAN, SNN, and CtIFT) are resource-demanding (Guo et al. 2024).
- *Limited Data:* Sparse or weakly annotated planetary image data sets limit training and validation for advanced matching algorithms (Geromichalos et al. 2020).

Table 8 highlights the limitations in feature matching for aerial and planetary imagery. Methods like 2chADCNN show vulnerability to seasonal and illumination variation, while GA-Net delivers high-quality disparity maps but with computational overhead unsuited to real-time applications. Joint descriptor methods achieve precise matching but depend heavily on diverse and well-aligned training data. These results emphasize the difficulty of achieving both generalizability and efficiency in feature matching for space and remote-sensing applications.

3.5.6 | Future Research Directions

1. *Efficient, lightweight deep models (e.g., CtIFT compression or quantization):* Pruning, quantization, or hardware accelerators to meet real-time flight constraints (Guo et al. 2024).
2. *Robustness and Generalization:* Leveraging synthetic data, domain adaptation, and sensor fusion to handle wide-ranging conditions (Naguib 2018).
3. *Hybrid approaches combining 2D–3D matching:* Integrating classical robust estimators (RANSAC and ICP) with learned features for adaptability and interpretability (Hwangbo et al. 2009).

TABLE 8 | (Feature matching): Quantitative illustration of persistent limitations.

Limitation	Representative method/data set	Matching accuracy (%)	Observation/constraint	Cited paper(s)
Environmental variability	2chADCNN for season-changing UAV-satellite matching	Drop in accuracy under snow/vegetation	Illumination and seasonal changes reduce the reliability of traditional features	Ren et al. (2023)
Computational complexity	GA-Net deep stereo matcher for remote sensing	High accuracy, low texture error	Computationally heavy, not real-time onboard suitable	Xia et al. (2020)
Limited data	Feature matching across satellite-aerial imagery with joint descriptors	~2 pixel match accuracy	Matching performance depends on viewpoint + training data diversity	Xu et al. (2022)

Abbreviations: GA-Net, geographic atrophy network; UAV, unmanned aerial vehicle.

4. *Particle filter integration*: Embedding matching steps within MCL systematically explores pose hypotheses—especially beneficial in data-sparse or rapidly changing planetary environs (Franchi and Ntagiou 2021).
5. *Multimodal integration* (vision + LiDAR + IMU) for real-time autonomous navigation (Van Pham et al. 2003).

By integrating classical geometry-based alignment with modern deep-learning and 3D modeling techniques, feature matching continues to enable resilient localization and mapping for spacecraft rendezvous, rover navigation, and in-orbit servicing.

4 | Evaluation of Space Image Data Sets

Evaluating available image data sets is crucial for advancing and deploying AI-enabled CV techniques in space missions. The choice and quality of data sets significantly influence training, validation, and the ultimate robustness of CV algorithms. While many data sets originated in earlier, classical CV-focused projects, they remain highly valuable for modern deep-learning research, especially with the addition of newly created synthetic data and cross-mission collaborations (Castano et al. 2003; Proenca and Gao 2020; Kisantal et al. 2020).

While data sets such as AI4Mars (Swan 2021) and LabelMars (Schwenzer et al. 2019) have significantly advanced supervised training for terrain segmentation and object detection, their limited size and class diversity still constrain model generalization. LabelMars, for example, provides only about 5000 annotated surface images—insufficient for training large deep models without overfitting. Recent studies suggest several mitigation strategies for data scarcity in planetary CV tasks.

Few-shot (Snell et al. 2017) and meta-learning (H. Li, Cui, et al. 2021) approaches enable models to generalize from limited labeled samples by leveraging prior knowledge across related domains. For instance, prototypical networks and metric-based meta-learning frameworks have demonstrated the ability to classify new terrain or rock types using only a handful of annotated examples. Transfer learning and self-supervised pretraining (e.g., contrastive and distillation methods) can also exploit vast unlabeled archives of rover or orbital imagery to build strong feature representations before fine-tuning on small planetary data sets. Additionally, domain randomization and synthetic-to-real transfer using high-fidelity simulated imagery (e.g. ESA PRL Martian Terrain Dataset or FDL Lunar DEMs) can augment data set diversity and reduce dependence on expensive manual labeling.

Integrating these techniques—especially few-shot adaptation and synthetic data augmentation—can mitigate the inherent data scarcity of planetary CV, enabling more robust and adaptive models for navigation, terrain understanding, and scientific exploration even under limited real imagery.

Below, we provide two tables summarizing (1) orbital and (2) planetary surface data sets, including their general descriptions, common CV-related applications, and known limitations. The first table focuses on data sets collected from orbiters (e.g., OSIRIS-REx or OREx (Origins, Spectral Interpretation, Resource Identification, and Security – Regolith Explorer) and Rosetta), while the second covers rovers and field trials in planetary analog environments. Together, they offer a broad

overview of what is currently available for researchers looking to develop and validate AI-based vision algorithms (Tables 9 and 10).

4.1 | Strategies for Data Set Utilization and Annotation

To address the challenges of limited annotations and improve the data set utility, the following approaches are proposed:

1. *Automated Annotation via Transfer Learning*: Pretrained models on labeled data sets (e.g., DeepMoon for crater detection) can generate pseudo-labels for unannotated orbital data sets like OREx images or Rosetta images. Weak supervision frameworks, such as Snorkel (Ratner et al. 2017), allow combining heuristic rules (e.g., geometric priors for crater shapes) with model predictions to refine labels efficiently. Recent work by Alshehhi and Gebhardt (2022) demonstrated that transfer learning from lunar data sets to Martian terrain improved the performance.
2. *Semisupervised Learning (SSL) for Planetary Data*: SSL methods like FlexMatch (B. Zhang, Wang, et al. 2021) and CoMatch (J. Li, Xiong, et al. 2021) leverage unlabeled data (e.g. [ESA PRL Martian Terrain Dataset](#)) by enforcing consistency between augmented views of the same image. For instance, FixMatch (Sohn et al. 2020) achieved 85% terrain-classification accuracy on LabelMars using only 10% labeled data, outperforming fully supervised baselines. This is critical for data sets like Erfoud, where manual labeling is impractical due to the data set size.
3. *Synthetic-to-Real Domain Adaptation*: GANs and style transfer techniques (Huang and Belongie 2017) can bridge the realism gap in synthetic data sets (e.g., SPEED or Artificial Lunar Landscape).
4. *MultiModal Active Learning*: Active learning frameworks (Settles 2009) prioritize labeling ambiguous samples (e.g., occluded craters in Moon Crater Database v1) by combining uncertainty sampling with multimodal data (LiDAR from ESA KADWijk Beach and images from ROBEX). This reduces annotation effort while maximizing model performance.
5. *Collaborative Annotation Platforms*: Federated learning systems (McMahan et al. 2017) enable distributed annotation of data sets like the Mars Crater Catalog across institutions without sharing raw data. Platforms like Zooniverse (Christian et al. 2012) have successfully crowdsourced labels for planetary science projects, ensuring scalability and diversity.

4.2 | Persistent Limitations

1. *Insufficient Realism and Diversity*: Even advanced synthetic data sets (e.g., SPEED and Artificial Lunar Landscape) often fail to capture the complexity of lighting, atmospheric, and geological conditions in real missions, limiting domain transfer. This gap persists despite advances in GANs and style transfer (Huang and

Belongie 2017), highlighting the need for more nuanced simulations.

2. *Label Scarcity*: Many orbital and planetary data sets (e.g., OREx images and Erfoud) remain unlabeled or sparsely annotated, hindering supervised training. While SSL methods like FixMatch (Sohn et al. 2020) mitigate this, their efficacy depends on the availability of some labeled data, which remains a bottleneck.
3. *Narrow Mission Scenarios*: Data sets often reflect specific mission contexts (e.g., ExoMars and PRISMA), reducing cross-domain generalizability.
4. *Dependency on Pretrained Models*: Automated annotation strategies relying on transfer learning (Ratner et al. 2017; Alshehhi and Gebhardt 2022) risk propagating biases from source domains (e.g., Earth or Moon) to target missions (e.g., Mars), limiting algorithmic robustness.

4.3 | Future Research Directions

1. *Enhanced Synthetic Data Generation*: Expand realism using physics-informed GANs and domain randomization to simulate extreme planetary conditions (e.g., Martian dust storms). Integrate multimodal synthetic data (e.g., LiDAR + thermal) to improve pretraining for data sets, like, FDL Localization Data set.
2. *Large-Scale Collaborative Annotation*: Develop federated learning frameworks (McMahan et al. 2017) and crowdsourced platforms (e.g., Zooniverse, Christian et al. 2012) to label planetary data sets (e.g., Mars Crater Catalog). Standardize annotation protocols to ensure consistency across agencies.
3. *Generalizable SSL*: Advance SSL methods like CoMatch (J. Li, Xiong, et al. 2021) to leverage unlabeled data from missions like [ESA PRL Martian Terrain Dataset](#), reducing reliance on manual labels. Prioritize uncertainty-aware active learning (Settles 2009) for ambiguous samples (e.g., occluded craters).
4. *MultiModal Fusion*: Integrate radar, LiDAR, and thermal data with vision-based data sets (e.g., ROBEX) to enhance redundancy and generalization in harsh environments.

By systematically compiling, annotating, and refining these orbital and planetary surface data sets—and supplementing them with realistic synthetic collections—researchers can enhance CV algorithm reliability, accelerate deep-learning progress, and better ensure the success of future space missions.

5 | Shortcomings and Future Prospects

Despite substantial progress in AI-driven CV for space robotics, multiple barriers continue to impede broader applicability and full operational autonomy. These shortcomings broadly relate to the fundamental constraints of onboard hardware, limited data set availability, generalization challenges, environmental complexity, and the necessity for robust algorithmic adaptability.

TABLE 9 | Orbital data sets.

Data set	Contents summary	Suitable uses	Limitations
OREx images	A large set of real images from multiple cameras and at different trajectories. Metadata available	– Landmark detection – Landing site localization – Hazard detection	No labels
Rosetta images (Keller et al. 2007)	A large set of real images from multiple cameras and at different trajectories. Estimated distance to the target available	– Landmark detection – Landing site localization – Hazard detection	No labels. Poor metadata available
DeepMoon (Silburt et al. 2019)	Data set of Moon craters used for training of CNN, already split into training, testing, and validation data sets. Each data set contains 30,000 DEM images, and the median number of craters per DEM image is 21. Ground truths available from crater catalogs. Images were generated by randomly cropping DEM images from LRO and Kaguya merged DEM	– Landmark detection – Landing site localization – Hazard detection	Lack of corresponding DEM and camera images
LROC NAC Images (Fairweather et al. 2022)	Data set of Moon craters based on LROC-NAC imagery. Training set contains 248 tiles from 16 NAC images where the craters have been marked manually with 43,402 craters	– Crater detection – Landing site localization – Hazard detection	No DEM data
Moon Crater Database v1 (Robbins 2019)	Contains approximately 1.3 million lunar craters > 1 km in diameter On the basis of LROC, LOLA, and SELENE Kaguya Terrain Camera images	– Crater detection – Landing site localization – Hazard detection	It is just a ground truth catalog, corresponding images have to be downloaded and matched separately
Mars Crater Catalog v1 Robbins (Robbins and Hynek 2012)	Contains approximately 640,000 Mars craters, about 385,000 of them with a diameter > 1 km On the basis of THEMIS Daytime IR planet-wide mosaic	– Crater detection – Landing site localization – Hazard detection	It is just a ground truth catalog, corresponding images have to be downloaded and matched separately
SPEED (Sharma et al. 2022)	A combination of synthetic and real images of Twin Anthropogenic Greenhouse gas Observers spacecraft from the PREcursore IperSpettrale della Missione Applicativa mission Contains 14,998 synthetic and 305 real images. Ground-truth pose labels, of translation vector and quaternion, are available	Pose estimation of an uncooperative target	Imagery of a single spacecraft only, which makes the trained algorithm dedicated to a single application

Abbreviations: CNN, Convolutional Neural Network; DEM, Digital Elevation Model; IR, infrared; LOLA, Lunar Orbiter Laser Altimeter; LRO, Lunar Reconnaissance Orbiter; LROC, Lunar Reconnaissance Orbiter Camera; NAC, Narrow Angle Camera; SELENE, Selenological and Engineering Explorer; SPEED, Spacecraft Pose Estimation Dataset; THEMIS, Thermal Emission Imaging System.

5.1 | Persistent Limitations

1. Computational Constraints

- *Onboard-resource limits:* Many state-of-the-art deep-learning models (e.g., large CNNs) remain too computationally heavy for space-qualified processors. With

tight power budgets and radiation-hardening requirements, resource utilization must be minimized.

- *Real-time demands:* Critical phases (e.g., EDL or close-proximity maneuvers around small bodies) require near-instantaneous inference. Complex deep-learning

TABLE 10 | Planetary surface data sets.

Data sets	Contents summary	Suitable uses	Limitations
Devon Island (Furgale et al. 2012)	Long traverse in a representative planetary environment, contains: – Nav camera stereo images – Sun vectors, inclinometer data – dGPS rover ground truth – 3D laser scans, image panoramas, and Digital Elevation Models (DEMs) (inc. GPS ground control points)	– Vision-based localization, navigation, and mapping – Object detection/segmentation	– No labels for training object classification-detection/segmentation AI models
ESA Seeker/SAFER (Paar et al. 2013)	ExoMars-like rover data products: – Navigation camera (Navcam), PanCam, and CLUPI images and IMU data – dGPS rover ground truth – Georeferenced DEMs and raw image products (w. GCPs)	– Vision-based localization, navigation, and mapping – Limited science-based scenarios, for example, using CLUPI images – Object detection/segmentation	– No labels for training object classification-detection/segmentation AI models – Some gaps are known to exist with the dGPS rover ground truth during some traverses
ESA ExoFIT (Balme et al. 2019)	ExoMars-like rover data products: – Navcam, PanCam, ISEM, and CLUPI images and IMU data – dGPS rover ground truth – Georeferenced DEMs and raw image products (w. GCPs)	– Vision-based localization, navigation, and mapping – Limited science-based scenarios, for example, using CLUPI images and ISEM data – Object detection/segmentation	– No labels for training object classification-detection/segmentation AI models – Unknown as to the specific contents, trajectories, or any other limitations
ESA PRL Martian Terrain Dataset	Collection of traverses at various lengths, totaling 13 km, in a representative planetary landscape, contains: – Multiple stereo camera streams: LocCam, Hazard Camera (HazCam), and PanCam – ToF camera – IMU data, wheel, and PTU odometry – LiDAR scans – dGPS rover ground truth – Georeferenced DEMs	– Vision-based localization, navigation, and mapping – Object detection/segmentation	– No labels for training object classification-detection/segmentation AI models
ESA Kadwijk Beach (Hewitt et al. 2018)	Collection of traverses at various lengths along the	– Vision-based localization,	– No labels for training object classification-

(Continues)

TABLE 10 | (Continued)

Data sets	Contents summary	Suitable uses	Limitations
	beach with artificial boulders, representing a planetary landscape, contains: – Multiple stereo camera streams: LocCam and PanCam – ToF camera – IMU data, wheel, and PTU odometry – LiDAR scans – dGPS rover ground truth – Georeferenced DEMs	navigation, and mapping – Object Detection/Segmentation	detection/segmentation AI models
Erfoud (Lacroix et al. 2020)	Long traverse in a representative planetary environment, contains: – Multiple stereo camera streams: navcam, front + rear hazcams – Multiple IMUs – LiDAR scans – dGPS rover ground truth – Georeferenced DEMs	– Vision-based localization, navigation, and mapping – Object detection/segmentation	– No labels for training object classification-detection/segmentation AI models – Data set documentation is incomplete, and some issues relating to calibration are known, solved, but yet to be released
MADMAX (Meyer et al. 2021)	A series of traverses of various lengths (up to 1.5 km) in a representative planetary environment, contains: – Mono stereo camera images – Color camera images – Omnidirectional camera – IMU data – dGPS rover ground truth	– Vision-based localization and navigation – Object detection/segmentation	– No DEM products – Frame drops of varying severity are present in all traverses – Camera decalibration in one set of traverses
Long Range Navigation Tests (ROBEX) (Verband Deutscher Maschinen- und Anlagenbau and Informationstechnische Gesellschaft 2018)	Long traverse in a representative planetary environment, contains: – Stereo camera images and depth image – IMU, wheel encoders and PTU position – dGPS rover ground truth – Georeferenced DEMs	– Vision-based localization and navigation – Object detection/segmentation	– No DEM products
LabelMars (Schwenzer et al. 2019)	– Collection of labeled Mars rover images for terrain classification	– Terrain classification – Rock segmentation/characterization – Planetary science/geological assessment	– Only ~ 5000 labeled images

(Continues)

TABLE 10 | (Continued)

Data sets	Contents summary	Suitable uses	Limitations
AI4Mars (Swan 2021)	Large collection (~35k) labeled Mars rover images for terrain classification, includes ~1.5k validation images	– Terrain classification – Rock segmentation/characterization – Planetary science/geological assessment	
Pessia et al. (2019)	9766 realistic renderings of various lunar terrain scenes, from a surface rover perspective Includes ground truth segmentation mask, discriminating small rock, big rock, sky, and “everything else” or surface classes. Bounding boxes of “large” rock locations also provided A small number of PCAM and TCAM images, with corresponding ground truth assets, are also provided	– Rock segmentation/characterization	– License conditions (TBC) – Synthetic images are ideal, but require image augmentation to improve realism
NASA surface image labeled data set (Stanboli and Wagstaff 2017)	Collection of ~7k images between 24 classes collected by MSL Mastcam Left and Right and MAHLI instruments to train image archive search	– Image context search	Classes are more relevant for image archive search applications rather than rover operations use cases
NASA Frontier Development Lab Localization Data set (Chung et al. 2018b)	A large collection of synthetic image panoramas to support the training and testing of localization methods, based specifically on Deep Learning, Includes a tool based on Unreal Engine 4 and the Moon Landscape v3.0 asset to create thousands of images.	– Vision-based localization	– Moon Landscape v3.0 is a paid-for asset, but it is low-cost

Abbreviations: 3D, three-dimensional; AI, Artificial Intelligence; CLUPI, close-up imager; dGPS, differential global positioning system; ESA, European Space Agency; ExoFIT, ExoMars-Like Field Trials; GCPs, ground control points; GPS, global positioning system; IMU, inertial measurement unit; ISEM, International Society for Ecological Modeling; LiDAR, Light Detection and Ranging; MADMAX, Morocco-Acquired data set of Mars-Analog eXploration; MSL, Mars Science Laboratory; PCAM, panoramic camera; PTU, Pan-Tilt-Unit; TCAM, terrain camera; ToF, time-of-flight.

architectures often must be downsampled or simplified, risking accuracy.

2. Data Scarcity and Variability

- *Limited labeled data sets*: While there are valuable orbital (e.g., LROC NAC and OSIRIS-REx) and surface (e.g., AI4Mars) image collections, many lack sufficient annotations for supervised deep learning. Specialized tasks like crater-based landmark navigation or rock segmentation require further labeling or customized data generation.
- *Narrow domain coverage*: Existing data sets typically reflect specific mission scenarios (ExoMars, PRISMA, etc.), reducing cross-environment generalizability. Even synthetic data sets (DeepMoon and NASA FDL) often lack the full realism required to handle extreme lighting or morphological complexities.

3. Environmental Challenges

- *Harsh conditions*: Extreme temperatures, cosmic radiation, dust storms, and irregular lighting degrade camera performance and can reduce the reliability of CV pipelines.
- *Unpredictable terrains*: Mars, lunar, or asteroid/regolith surfaces lack consistent features and often have dynamic or dust-laden environments that hamper stable feature detection and pose estimation.

4. Generalization and Robustness

- *Domain shifts*: Models tailored to one planetary environment may fail when transferred to another with distinct topographic or visual properties (e.g., from lunar crater detection to Martian dune fields).
- *Edge cases*: Uncooperative targets (e.g., noncooperative spacecraft and tumbling debris) or novel terrain features

can derail models trained under more controlled assumptions.

5. Integration and Adaptability

- *Stand-alone CV modules*: Many pipelines remain siloed from the broader GNC loop. Fusing multisensor data (radar, LiDAR, IMU) or shape modeling (e.g., stereo-photoclinometry) demands streamlined flight software architectures.
- *Lack of continuous learning*: Missions often rely on pretrained models with minimal capacity for onboard updates or adaptation to evolving mission conditions.

5.2 | Future Research Directions

1. Efficient Onboard Architectures

- *Lightweight model design*: Emphasize pruning, quantization, and specialized hardware accelerators (DSPs, FPGAs, low-power GPUs) for real-time inference within tight power budgets.
- *Neural architecture search (NAS)*: Algorithmic exploration to discover minimal yet high-performing CV architectures suitable for flight processors.

2. Enhanced Synthetic Data and Domain Adaptation

- *High-fidelity simulation*: Advanced GAN-based or physics-driven environments can capture extremes of lighting, dust, or morphological variety, bridging the gap between Earth-based training and planetary deployment.
- *Transfer learning*: Pretraining on terrestrial or synthetic data sets, followed by domain adaptation methods (e.g., unpaired image-to-image translation) to refine the model for actual mission imagery.

3. Adaptive and Resilient Algorithms

- *Onboard retraining or fine-tuning*: Allowing partial model updates in response to novel terrain or sensor degradation fosters real-time adaptability.
- *Robust pose estimation and detection*: Develop hybrid pipelines (deep learning + classical geometry) to mitigate sensitivity to illumination changes, occlusions, and reflective surfaces.

4. MultiSensor Fusion and Systems Integration

- *Tightly coupled sensor data*: Incorporate radar, LiDAR, inertial, and multispectral imaging to increase reliability when a single sensor is compromised (e.g., dusty camera lens).
- *Flight software alignment*: Integrate CV modules with GNC frameworks (e.g., Kalman filters and trajectory planners) to enable coordinated, closed-loop autonomy.

5. Continuous Onboard Learning

- *Incremental model improvement*: Investigate semi-supervised or reinforcement-based methods, leveraging newly acquired mission data to refine model weights.
- *Balancing risk versus reward*: Carefully manage in-mission learning protocols with safety constraints—particularly for critical mission phases (e.g., EDL and docking).

Addressing these limitations through cohesive, interdisciplinary research is vital for advancing the reliability, autonomy, and

scientific return of upcoming space missions. By concentrating on computational efficiency, expanded data set realism, adaptive and robust algorithms, and well-designed multisensor integration, AI-driven CV methods can substantially broaden mission capabilities, de-risk ambitious exploration targets, and propel space robotics toward greater autonomy and success.

6 | Conclusions

This paper has provided a comprehensive review of AI-enabled CV technologies that are essential for current and future space robotic missions. Emphasis was placed on key mission phases—EDL, Orbital Operations, and Planetary Surface Exploration—highlighting significant advancements in TRN, semantic segmentation, relative pose estimation, and feature matching.

Notably, DNN-based machine-learning techniques have demonstrated significant advantages across these diverse CV tasks, showcasing potential for enhancing mission autonomy and robustness. The paper also reviewed and summarized various space image data sets, crucial for realistic and effective training and benchmarking of DNN models, providing essential resources for ongoing research.

Nevertheless, challenges such as computational constraints, limited robustness and generalizability, and data scarcity continue to restrict the broader adoption and effectiveness of these advanced CV technologies in practical space missions. Future research must prioritize the development of computationally efficient DNN architectures, the enhancement of synthetic data realism, and the implementation of adaptive algorithms capable of real-time adjustments to dynamic conditions. Addressing these challenges will enable the next generation of autonomous and robust space robotic missions, significantly expanding the capabilities and success of planetary exploration.

Emerging solutions, such as radiation-hardened lightweight AI, physics-driven simulations, and federated learning frameworks, offer pathways to bridge robustness gaps. Multisensor fusion (e.g., vision with LiDAR) and explainable AI could further ensure operational safety in high-risk scenarios like docking or hazard avoidance. However, challenges like algorithmic bias in cross-domain generalization and validation in extreme environments demand rigorous interdisciplinary collaboration. By harmonizing these advances, AI-driven CV will catalyze breakthroughs in autonomous exploration, enabling ambitious endeavors, such as lunar lava tube mapping or Martian sample return, ultimately solidifying its role as the cornerstone of next-generation space robotics.

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Ethics Statement

The authors have nothing to report.

Consent

This article does not contain any studies with human or animal subjects performed by any of the authors.

Conflicts of Interest

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Data Availability Statement

The authors have nothing to report.

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